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- Overview of 'semantic vision'?
- Image classification
- Bag-of-words
 - Recall
 - Bag-of-feature
 - BOW for image classification
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 - K nearest neighbors
 - Naïve Bayes
 - Support vector machine
- Evaluation and hyperparameters selection



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Is this a streetlight?

(Recognition / classification)



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Where are the people?

(Detection)



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Is that Potala palace?

(Identification)



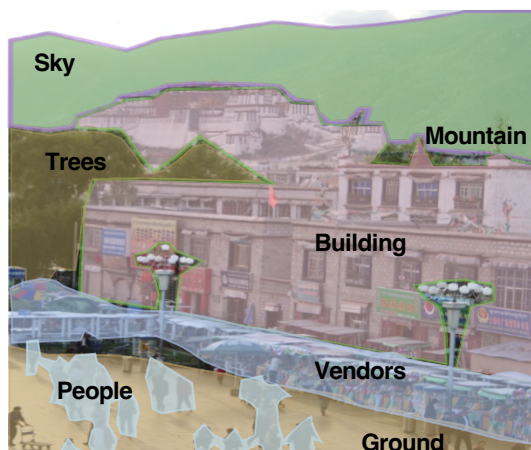
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What's in the scene?

(semantic segmentation)



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What type of scene is it?

Scene categorization



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What are these people doing?

Activity / Event Recognition

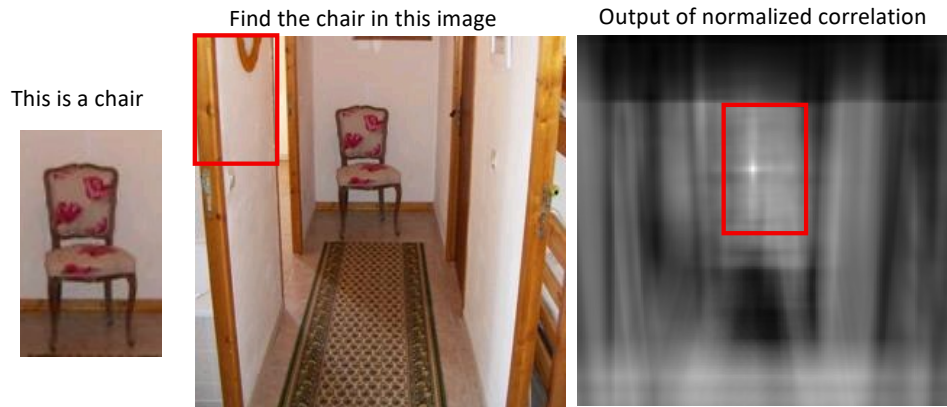


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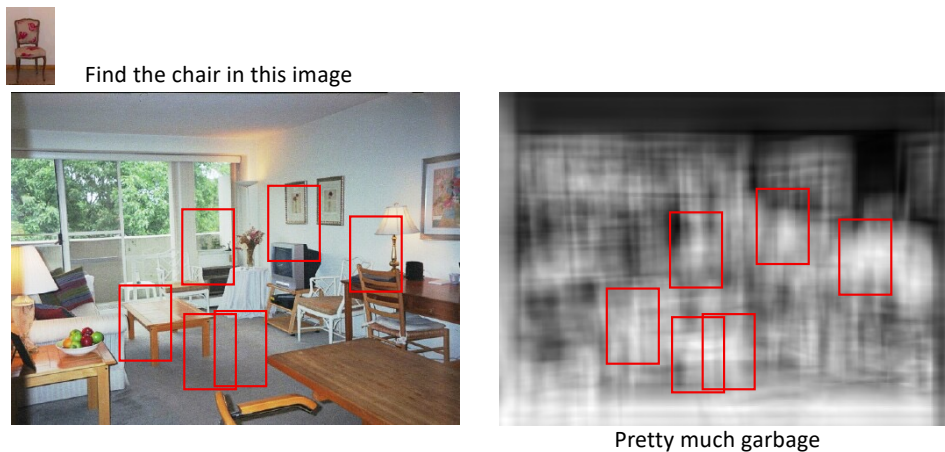
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Object recognition: Is it really so hard?



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Object recognition: Is it really so hard?



Pretty much garbage

Simple template matching is not going to make it
A "popular method is that of template matching, by point to point correlation of a model pattern with the image pattern. These techniques are inadequate for three-dimensional scene analysis for many reasons, such as occlusion, changes in viewing angle, and articulation of parts." Nivatia & Binford, 1977.

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And it can get a lot harder



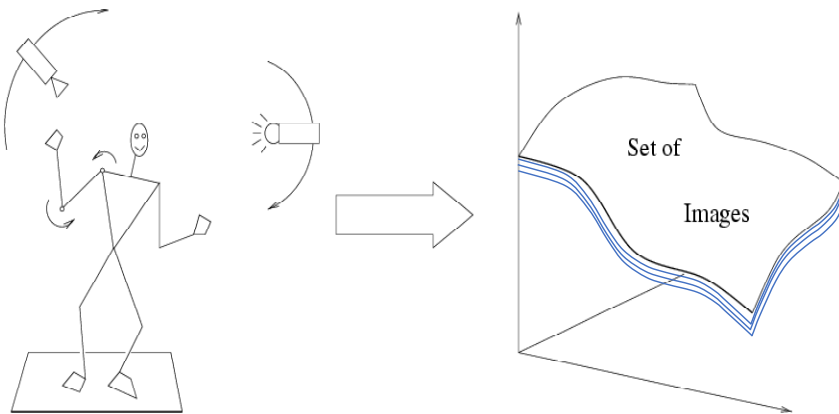
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Brady, M. J., & Kersten, D. (2003). Bootstrapped learning of novel objects. *J Vis*, 3(6), 413-422

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Why is this hard?



Variability: Camera position
Illumination
Shape parameters



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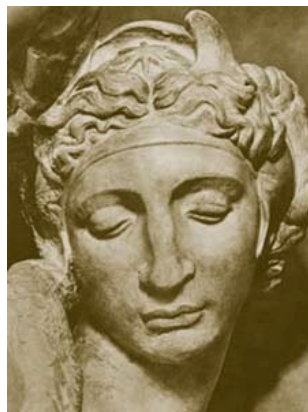
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Why is this hard?



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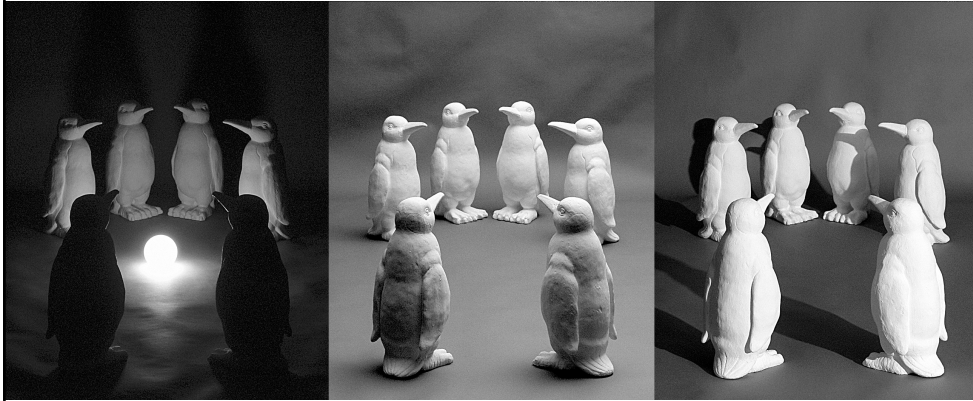
Challenge: variable viewpoint



Michelangelo 1475-1564

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Challenge: variable illumination



Challenge: scale

and small things
from Apple.
(Actual size)



Challenge: deformation

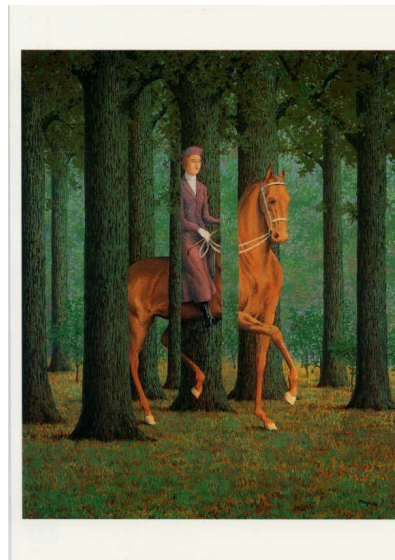


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Challenge: Occlusion



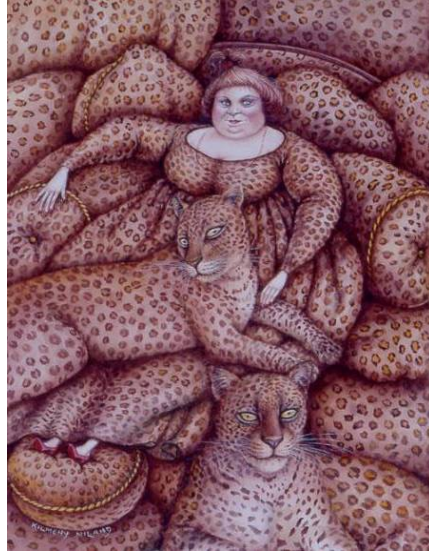
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Magritte, 1957

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Challenge: background clutter



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Challenge: background clutter



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Challenge: intra-class variations



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Svetlana Lazebnik

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Image Classification



(assume given set of discrete labels)
{dog, cat, truck, plane, ...}

→ cat



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Image Classification: Problem



08	02	22	97	38	15	00	40	00	75	04	05	07	78	52	12	50	77	17	49
49	49	99	40	17	81	18	57	60	87	17	40	99	43	69	47	74	56	62	00
81	49	31	73	55	79	14	29	93	71	40	67	54	85	30	03	49	13	36	65
52	70	95	23	04	60	11	92	13	87	65	56	01	32	56	71	37	02	36	91
22	31	56	71	55	67	45	41	92	36	54	22	40	40	28	66	33	13	80	
24	43	34	47	99	03	45	02	44	75	35	53	78	36	94	20	35	17	12	50
12	58	81	28	64	23	67	10	26	38	40	67	59	54	70	66	18	38	64	70
67	26	20	68	02	62	12	20	95	63	94	39	63	08	40	91	66	49	94	21
24	55	58	05	66	73	99	26	97	17	78	78	96	83	14	88	34	89	63	72
21	36	23	09	75	00	76	44	20	45	35	14	00	61	33	97	34	31	33	95
78	17	59	28	22	75	31	67	15	94	03	80	04	62	16	14	09	53	56	92
16	39	05	42	96	35	31	47	55	58	88	24	00	17	54	24	36	29	85	57
86	56	00	48	35	71	89	07	05	44	44	37	44	60	21	58	51	54	17	58
19	80	81	68	05	94	47	69	28	73	92	13	86	52	17	77	04	89	55	40
04	52	05	83	97	35	99	16	07	97	57	32	16	26	26	79	33	27	98	66
95	44	46	87	57	62	20	72	03	46	33	67	46	55	12	32	63	93	53	69
04	42	14	73	79	48	11	24	94	72	18	08	46	29	32	40	62	76	36	
20	69	36	41	72	30	23	03	74	45	69	82	67	59	85	74	04	36	16	
20	73	35	29	78	31	90	01	74	31	49	71	47	74	41	16	23	57	05	54
01	70	54	71	83	51	54	69	16	92	33	49	63	43	52	01	19	67	48	

What the computer sees

image classification

82% cat
15% dog
2% hat
1% mug



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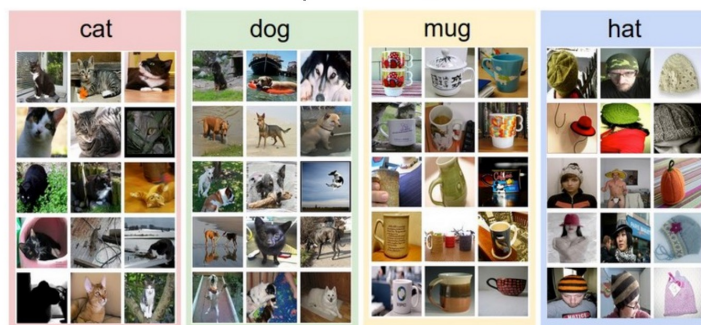
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Data-driven approach

- Collect a database of images with labels
- Use ML to train an image classifier
- Evaluate the classifier on test images

Example training set



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A simple pipeline - Training

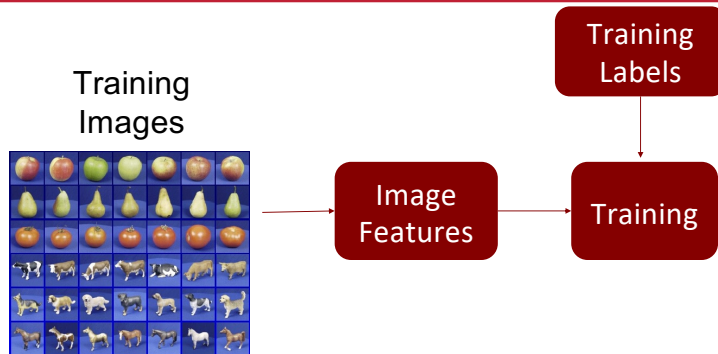
Training
Images



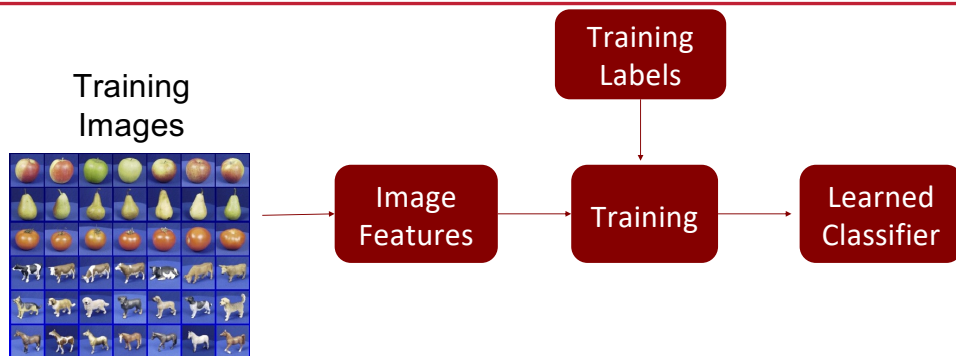
Image
Features

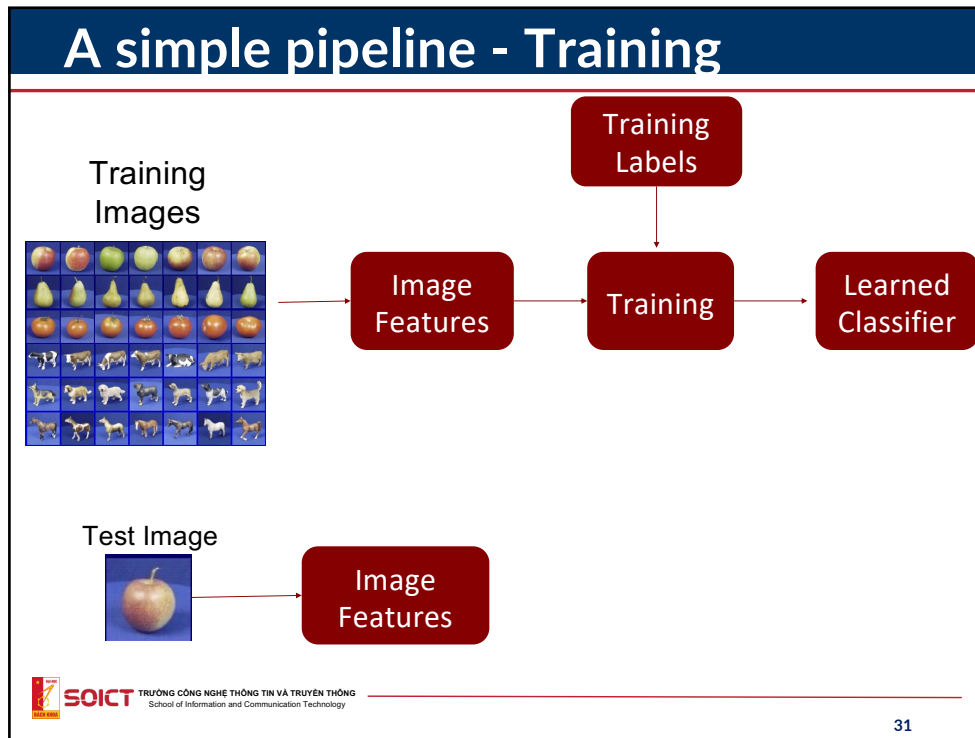
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A simple pipeline - Training

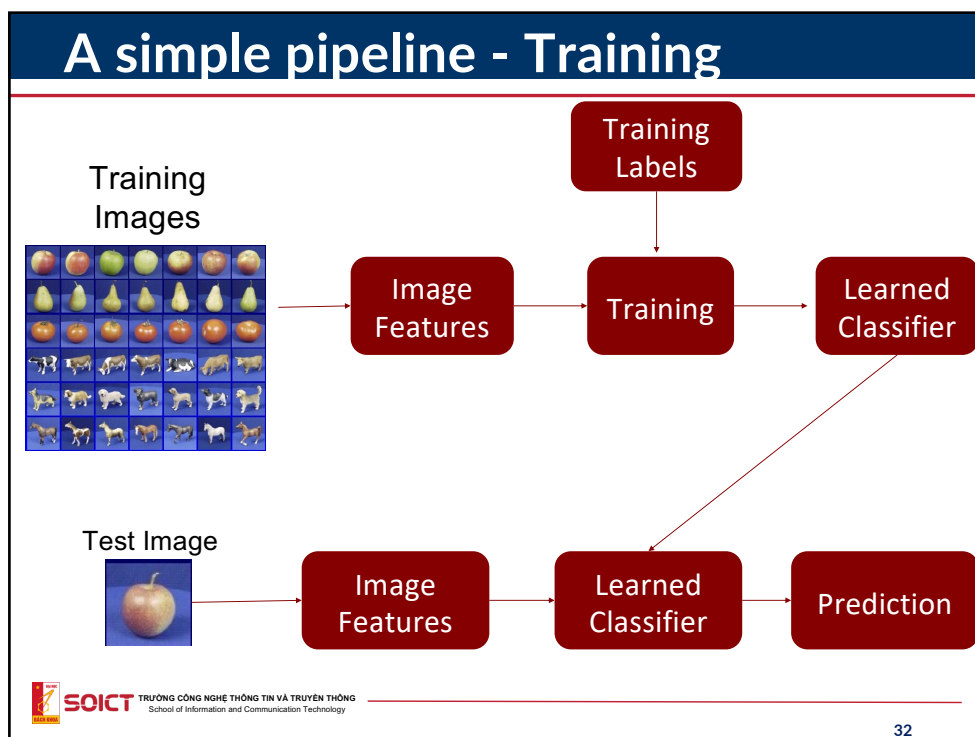


A simple pipeline - Training





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Trích chọn đặc trưng (nhắc lại)

- Hai loại đặc trưng được trích chọn từ ảnh:
 - Đặc trưng cục bộ và toàn cục
- **Đặc trưng toàn cục:**
 - Mô tả toàn bộ ảnh như 1 đối tượng
 - Đặc trưng đường biên, đặc trưng hình dạng, đặc trưng kết cấu
 - Ví dụ: Invariant Moments (Hu, Zernike), Histogram Oriented Gradients (HOG), PHOG, and Co-HOG,...
- **Đặc trưng cục bộ:**
 - Mô tả đặc trưng cục bộ mô tả từng vùng nhỏ trong ảnh, từng vùng cục bộ của đối tượng (điểm đặc trưng trong ảnh).
 - Biến đổi đặc trưng kết cấu/màu sắc trong mỗi vùng cục bộ ảnh
 - Ví dụ: SIFT, SURF, LBP, BRISK, MSER và FREAK, ...



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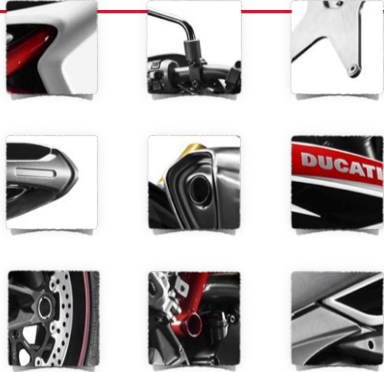
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Some local feature are very informative

An object as





a collection of local features
(bag-of-features)

- deals well with occlusion
- scale invariant
- rotation invariant

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CalTech6 dataset








class	bag of features	bag of features	Parts-and-shape model
	Zhang et al. (2005)	Willamowski et al. (2004)	Fergus et al. (2003)
airplanes	98.8	97.1	90.2
cars (rear)	98.3	98.6	90.3
cars (side)	95.0	87.3	88.5
faces	100	99.3	96.4
motorbikes	98.5	98.0	92.5
spotted cats	97.0	—	90.0

Works pretty well for image-level classification

Csurka et al. (2004), Willamowski et al. (2005), Grauman & Darrell (2005), Sivic et al. (2003, 2005)



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Bag-of-features

represent a data item (document, texture, image)
as a histogram over features

an old idea

(e.g., texture recognition and information retrieval)

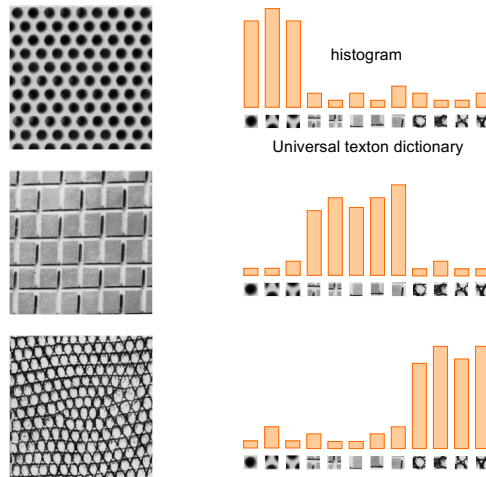


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Texture recognition

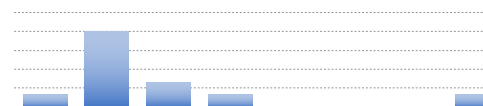


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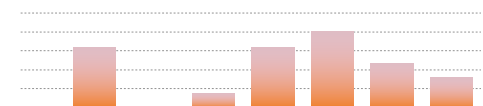
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Vector Space Model



1	6	2	1	0	0	0	1
Tartan	robot	CHIMP	CMU	bio	soft	ankle	sensor



0	4	0	1	4	5	3	2
Tartan	robot	CHIMP	CMU	bio	soft	ankle	sensor



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G. Salton, "Mathematical and Information Retrieval" Journal of Documentation 1979

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A document (datapoint) is a vector of counts over each word (feature)

$$\mathbf{v}_d = [n(w_{1,d}) \ n(w_{2,d}) \ \cdots \ n(w_{T,d})]$$

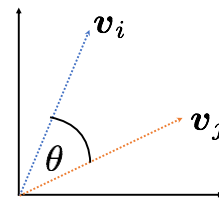
$n(\cdot)$ counts the number of occurrences

just a histogram over words

What is the similarity between two documents?

Use any distance you want but the cosine distance is fast.

$$\begin{aligned} d(\mathbf{v}_i, \mathbf{v}_j) &= \cos \theta \\ &= \frac{\mathbf{v}_i \cdot \mathbf{v}_j}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|} \end{aligned}$$

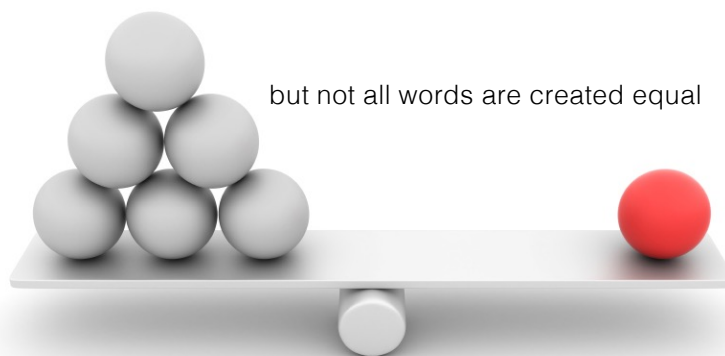


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Remember



but not all words are created equal



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TF-IDF

Term **F**requency Inverse **D**ocument **F**requency

$$\mathbf{v}_d = [n(w_{1,d}) \quad n(w_{2,d}) \quad \cdots \quad n(w_{T,d})]$$

weigh each word by a heuristic

$$\mathbf{v}_d = [n(w_{1,d})\alpha_1 \quad n(w_{2,d})\alpha_2 \quad \cdots \quad n(w_{T,d})\alpha_T]$$

$$n(w_{i,d})\alpha_i = n(w_{i,d}) \log \left\{ \frac{D}{\sum_{d'} \mathbf{1}[w_i \in d']} \right\}$$

(down-weights **common** terms)



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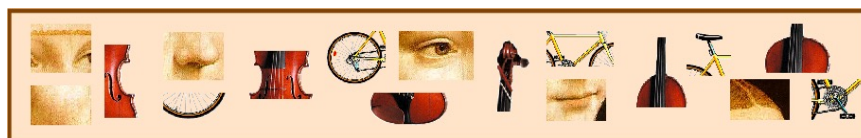
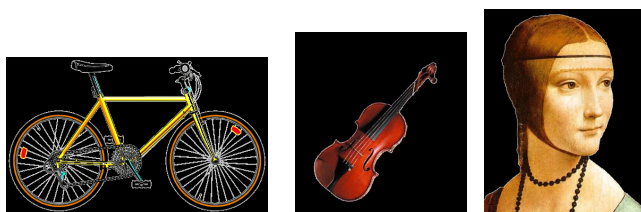
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Mô hình túi từ

- Đặc trưng cục bộ ~ 1 từ
- 1 ảnh ~ 1 văn bản
- Sử dụng mô hình vector: ảnh = tần suất xuất hiện của các từ trực quan



Standard BOW pipeline for image classification

Dictionary Learning:

Learn Visual Words using clustering

Encode:

build Bags-of-Words (BOW) vectors
for each image

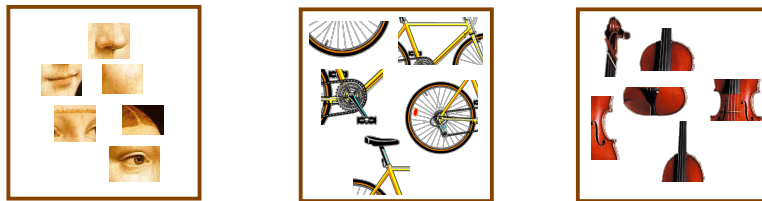
Classify:

Train and test data using BOWs

Standard BOW pipeline for image classification

Dictionary Learning:
Learn Visual Words using clustering

1. extract features (e.g., SIFT) from images



Standard BOW pipeline for image classification

Dictionary Learning:
Learn Visual Words using clustering

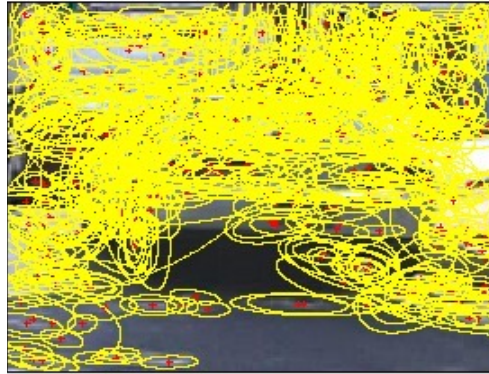
2. Learn visual dictionary (e.g., K-means clustering)



Standard BOW pipeline for image classification

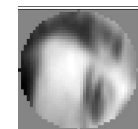
What kinds of features can we extract?

- Regular grid
 - Vogel & Schiele, 2003
 - Fei-Fei & Perona, 2005
- Interest point detector
 - Csurka et al. 2004
 - Fei-Fei & Perona, 2005
 - Sivic et al. 2005
- Other methods
 - Random sampling (Vidal-Naquet & Ullman, 2002)
 - Segmentation-based patches (Barnard et al. 2003)

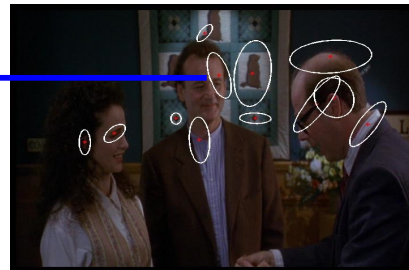


Standard BOW pipeline for image classification

Compute SIFT
descriptor
[Lowe'99]



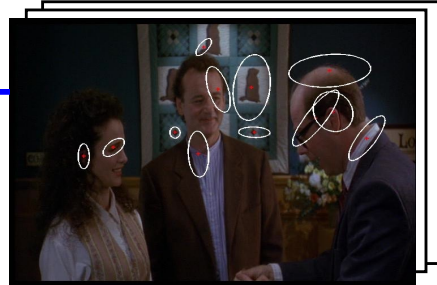
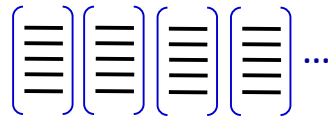
Normalize patch



Detect patches

[Mikojaczyk and Schmid '02]
[Mata, Chum, Urban & Pajdla, '02]
[Sivic & Zisserman, '03]

Standard BOW pipeline for image classification



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Standard BOW pipeline for image classification

How do we learn the dictionary?

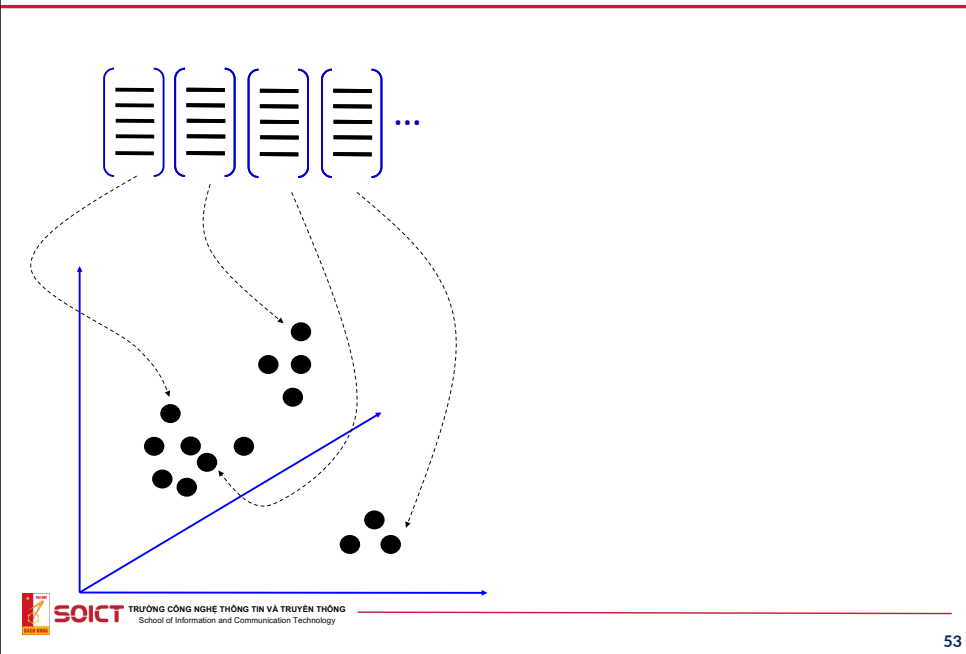


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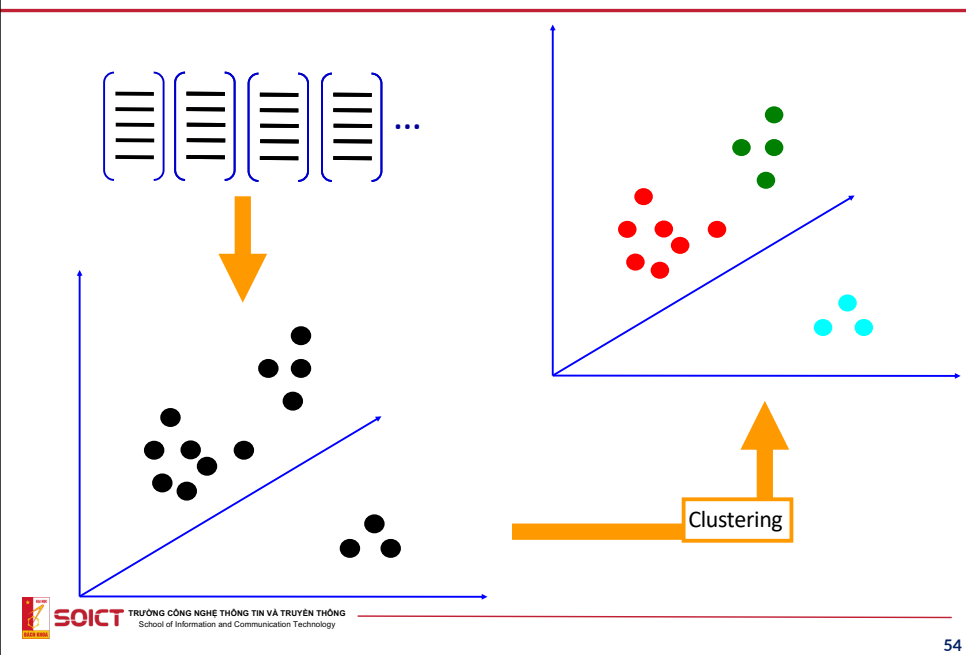
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Standard BOW pipeline for image classification

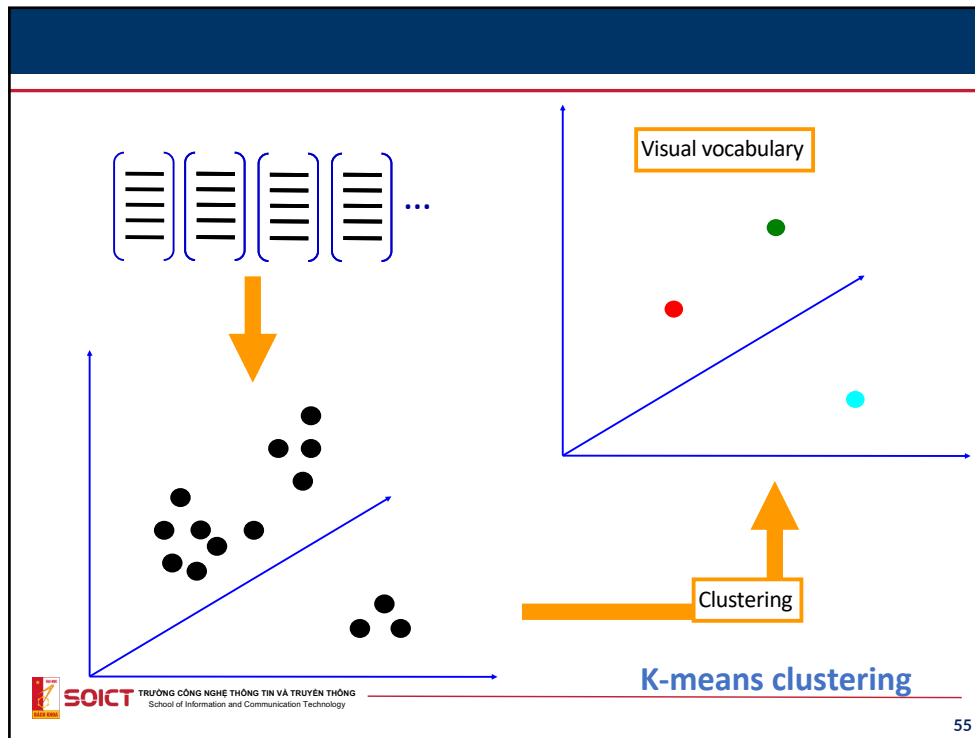


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Standard BOW pipeline for image classification



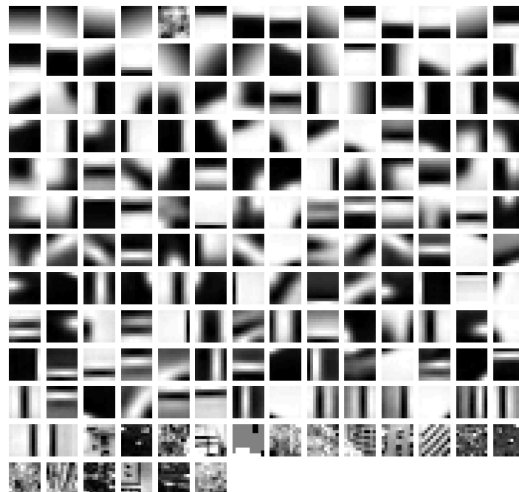
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Standard BOW pipeline for image classification

Example visual dictionary



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Standard BOW pipeline for image classification

Example visual dictionary



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Standard BOW pipeline for image classification

Another dictionary



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Standard BOW pipeline for image classification

Dictionary Learning:

Learn Visual Words using clustering

Encode:

build Bags-of-Words (BOW) vectors
for each image

Classify:

Train and test data using BOWs

Standard BOW pipeline for image classification



1. Quantization: image features gets associated to a visual word (nearest cluster center)



Encode:

build Bags-of-Words (BOW) vectors
for each image

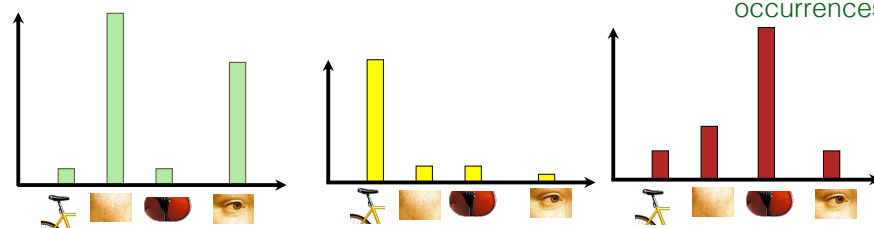


Standard BOW pipeline for image classification

Encode:

build Bags-of-Words (BOW) vectors
for each image

2. Histogram: count the
number of visual word
occurrences



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Standard BOW pipeline for image classification



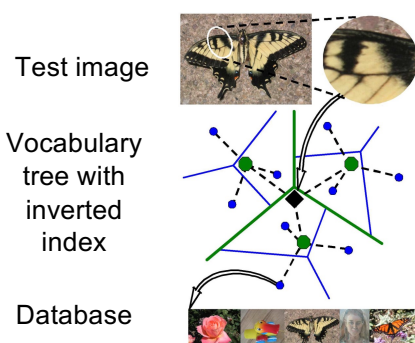
codewords

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Standard BOW pipeline for image classification

Scalability: Alignment to large databases

- What if we need to align a test image with thousands or millions of images in a model database?
 - Efficient putative match generation
 - Fast nearest neighbor search, inverted indexes



D. Nistér and H. Stewenius, [Scalable Recognition with a Vocabulary Tree](#), CVPR 2006

slide: S. Lazebnik

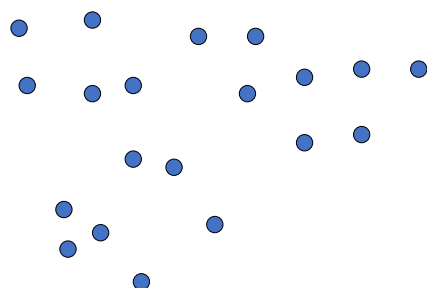


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What is a Vocabulary Tree?



Nistér and Stewenius CVPR 2006

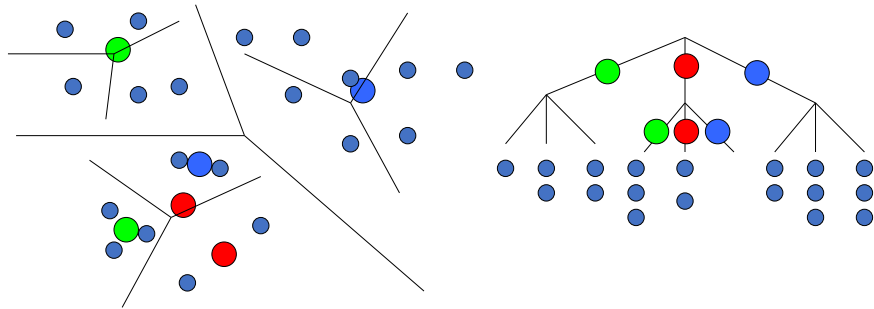


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What is a Vocabulary Tree?



- Multiple rounds of K-Means to compute decision tree (offline)
- Fill and query tree online



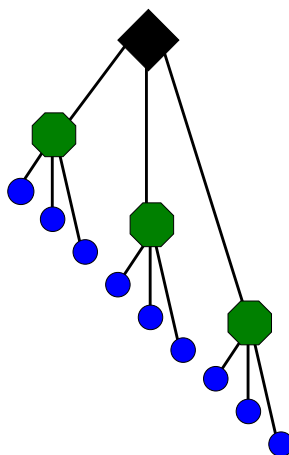
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Nister and Stewenius CVPR 2006

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Populating the vocabulary tree /inverted index



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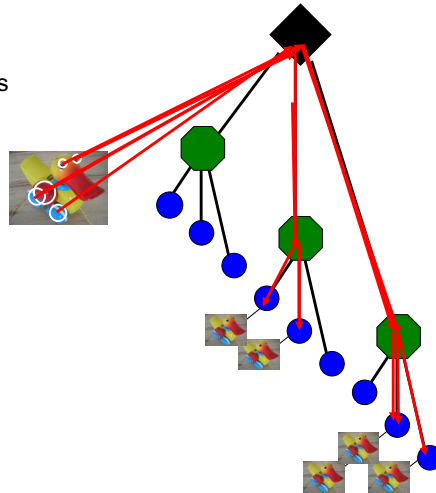
Slide credit: D. Nister

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Populating the vocabulary tree/inverted index

Model images

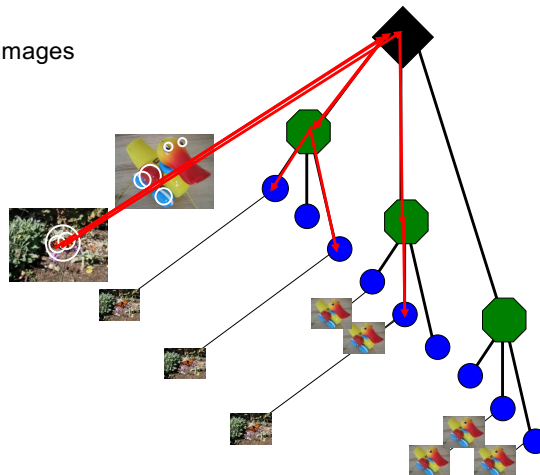


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Populating the vocabulary tree

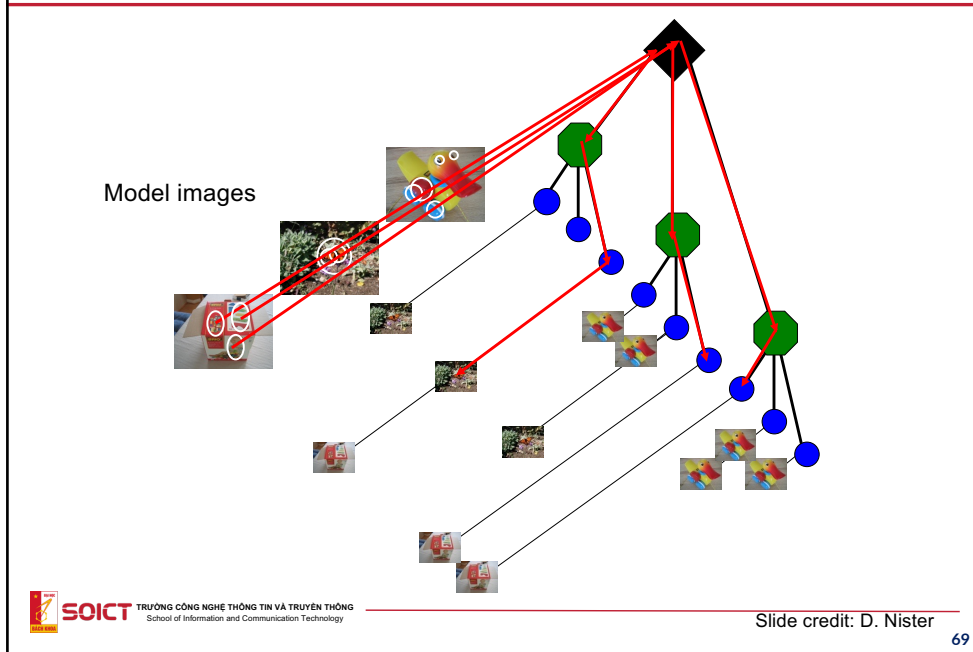
Model images



Slide credit: D. Nister

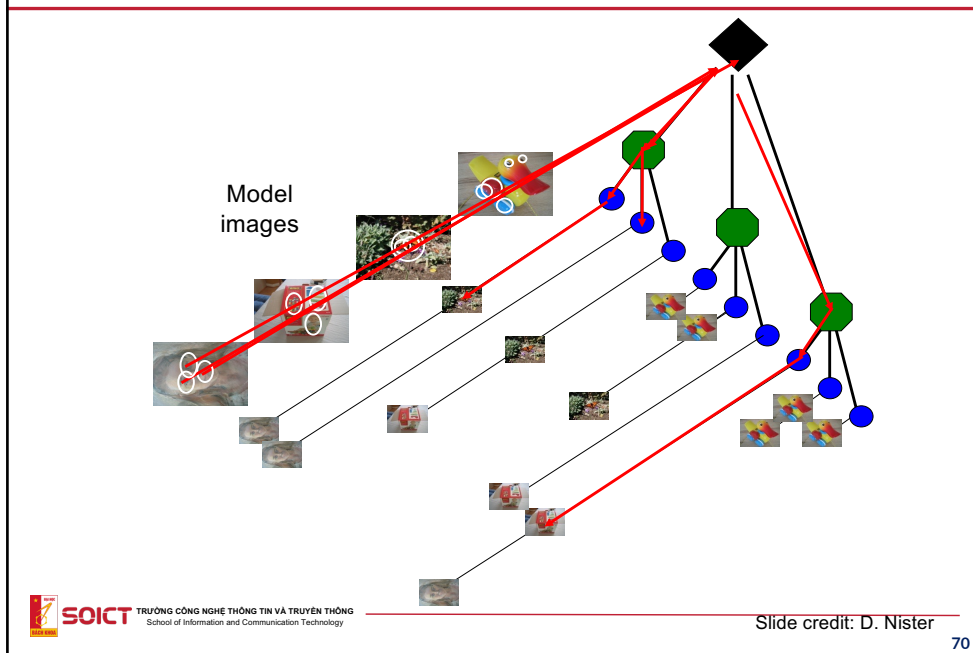
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Populating the vocabulary tree



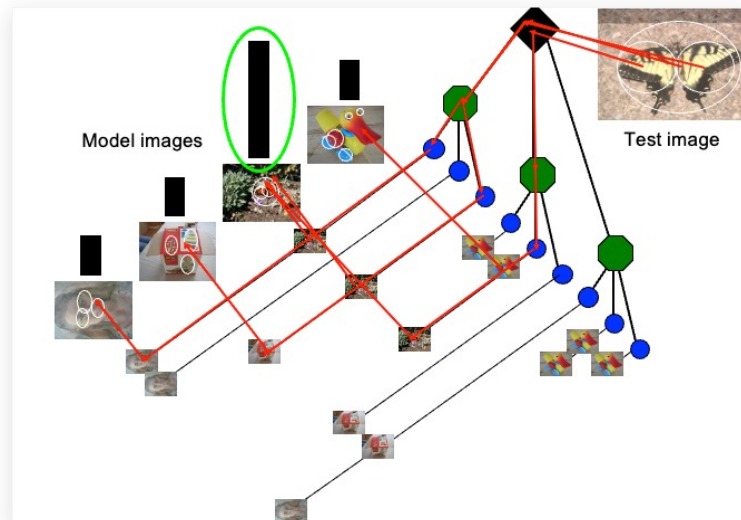
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Populating the vocabulary tree



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Looking up a test image



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Slide credit: D. Nister

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Standard BOW pipeline for image classification

Dictionary Learning:
Learn Visual Words using clustering

Encode:
build Bags-of-Words (BOW) vectors
for each image

Classify:
Train and test data using BOWs

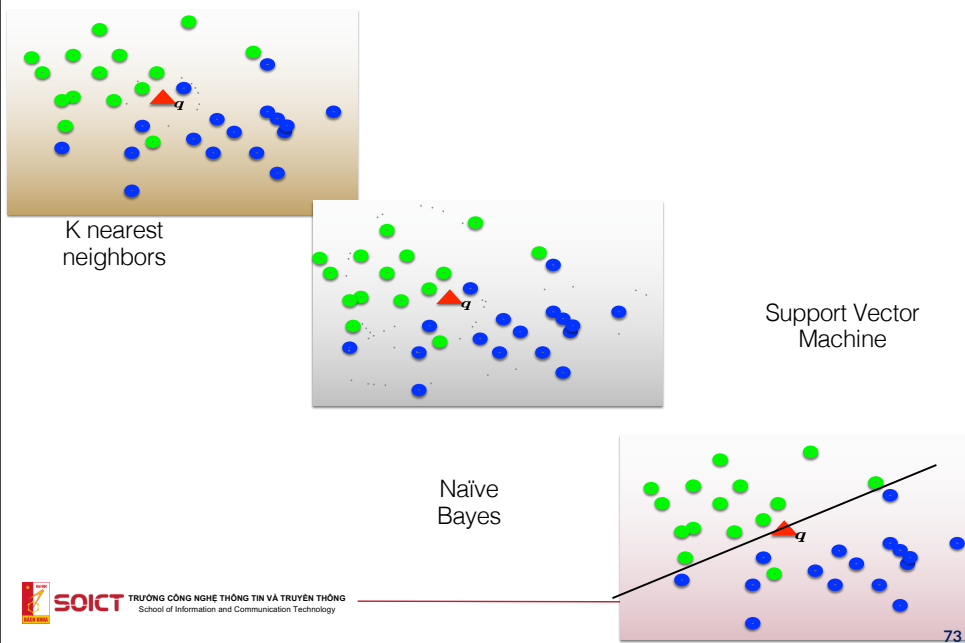


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Standard BOW pipeline for image classification



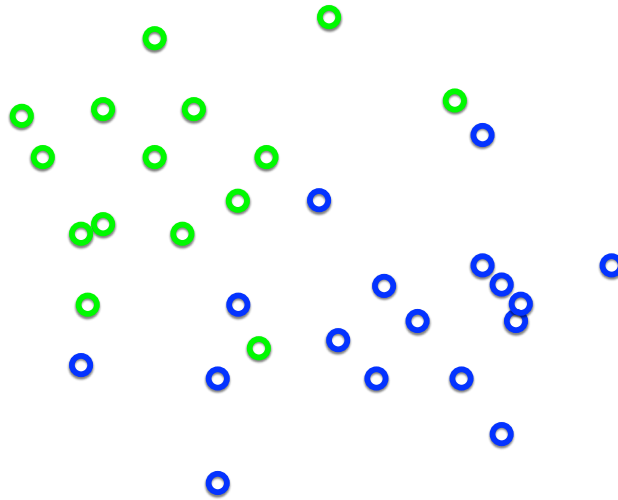
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Distribution of data from two classes

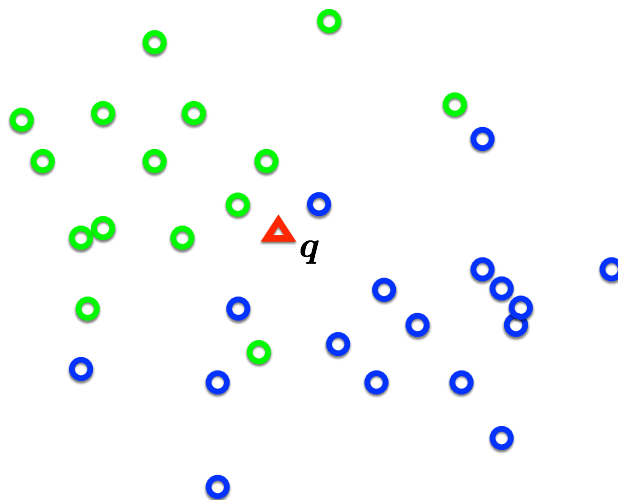


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Distribution of data from two classes



Which class does q belong too?

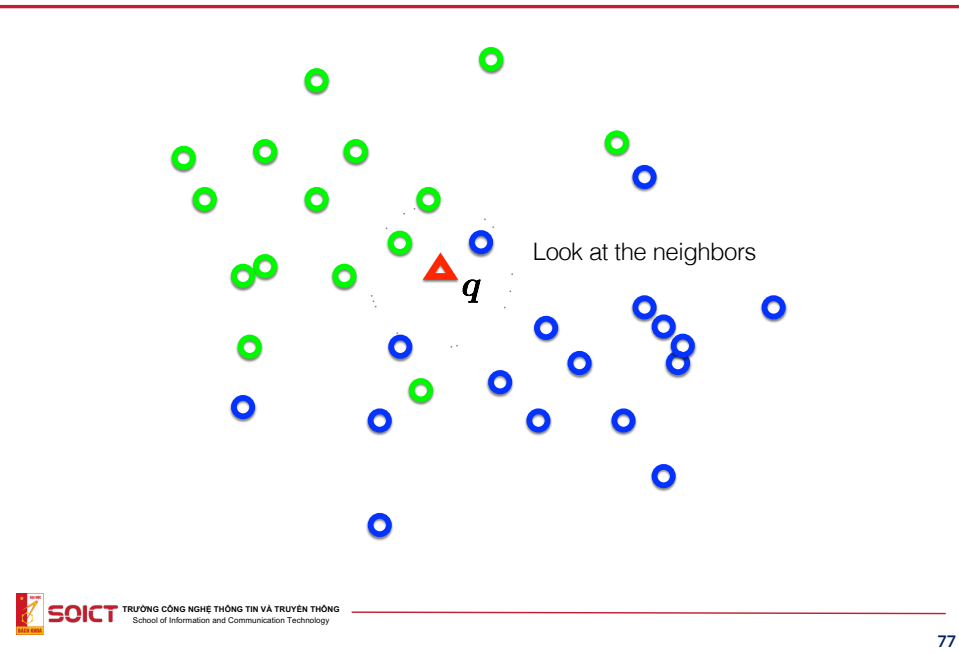


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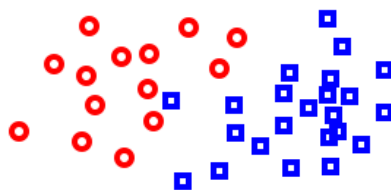
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Distribution of data from two classes



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K-Nearest Neighbor (KNN) Classifier

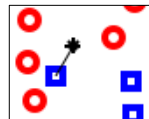


Non-parametric pattern classification approach

Consider a two class problem where each sample consists of two measurements (x,y) .

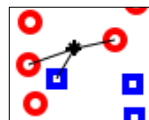
For a given query point q , assign the class of **the nearest neighbor**

$k = 1$



Compute the **k nearest neighbors** and assign the class by majority vote.

$k = 3$



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Nearest Neighbor is competitive

	Test Error Rate (%)
Linear classifier (1-layer NN)	12.0
K-nearest-neighbors, Euclidean	5.0
K-nearest-neighbors, Euclidean, deskewed	2.4
MNIST Digit Recognition	
– Handwritten digits	
– 28x28 pixel images: $d = 784$	
– 60,000 training samples	
– 10,000 test samples	
Yann LeCunn	
K-NN, Tangent Distance, 16x16	1.1
K-NN, shape context matching	0.67
1000 RBF + linear classifier	3.6
SVM deg 4 polynomial	1.1
2-layer NN, 300 hidden units	4.7
2-layer NN, 300 HU, [deskewing]	1.6
LeNet-5, [distortions]	0.8
Boosted LeNet-4, [distortions]	0.7



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Remark

What is the best distance metric between data points?

- Typically Euclidean distance
- Locality sensitive distance metrics
- Important to normalize.
Dimensions have different scales

How many K?

- Typically $k=1$ is good
- Cross-validation (try different k !)



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Distance metrics

$$D(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_N - y_N)^2} \quad \text{Euclidean}$$

$$D(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} = \frac{x_1 y_1 + \cdots + x_N y_N}{\sqrt{\sum_n x_n^2} \sqrt{\sum_n y_n^2}} \quad \text{Cosine}$$

$$D(\mathbf{x}, \mathbf{y}) = \frac{1}{2} \sum_n \frac{(x_n - y_n)^2}{(x_n + y_n)} \quad \text{Chi-squared}$$



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Distance metrics

L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

L2 (Euclidean) distance

$$d_2(I_1, I_2) = \sqrt{\sum_p (I_1^p - I_2^p)^2}$$

- Two most commonly used special cases of p-norm

$$\|\mathbf{x}\|_p = (|x_1|^p + \cdots + |x_n|^p)^{\frac{1}{p}} \quad p \geq 1, \mathbf{x} \in \mathbb{R}^n$$



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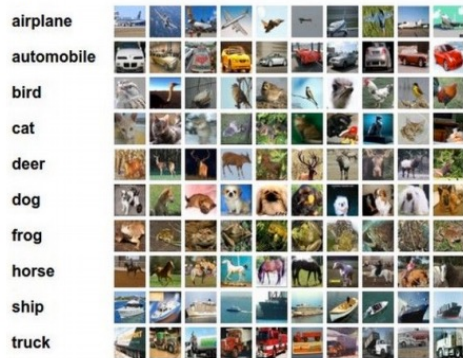
CIFAR-10 and NN results

Example dataset: **CIFAR-10**

10 labels

50,000 training images

10,000 test images.



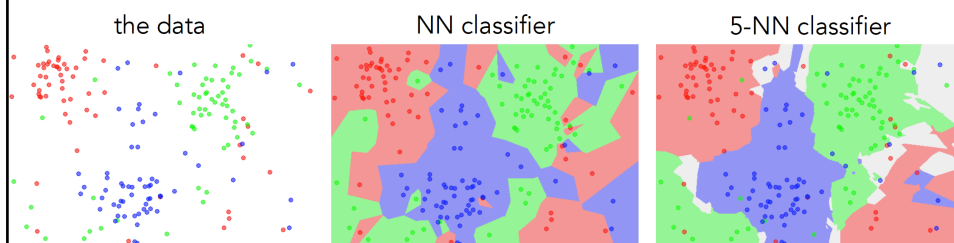
For every test image (first column),
examples of nearest neighbors in rows



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k-nearest neighbor

- Find the k closest points from training data
- Labels of the k points **"vote"** to classify



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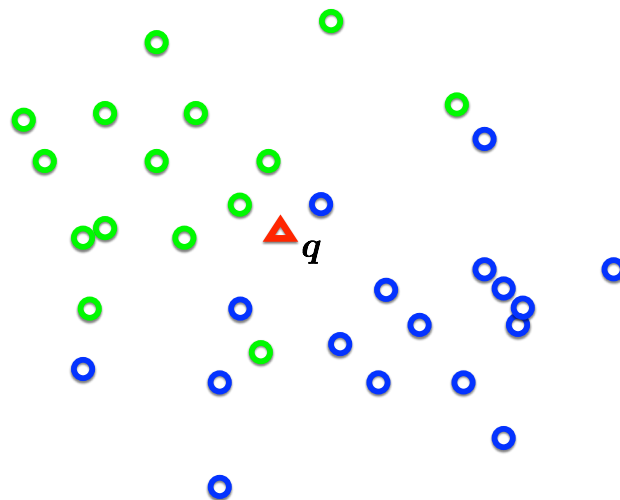


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Distribution of data from two classes



Which class does q belong too?

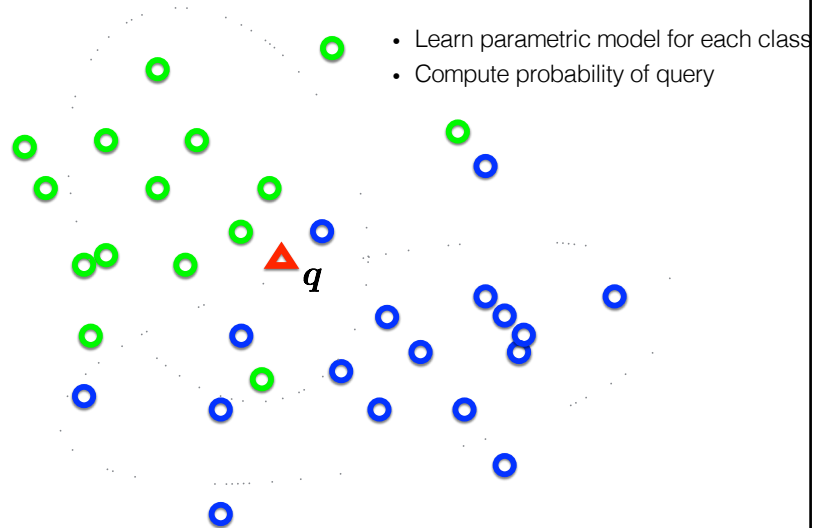


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Distribution of data from two classes



Posterior

This is called the posterior:
the probability of a class z given the observed features X

$$p(z|X)$$

For classification, z is a
discrete random
variable
(e.g., car, person, building)

X is a set of observed features
(e.g., features from a single image)

(it's a function that returns a single probability value)

Posterior

This is called the posterior:
the probability of a class z given the observed features X

$$p(z|x_1, \dots, x_N)$$

For classification, z is a
discrete random
variable
(e.g., car, person, building)

Each x is an observed feature
(e.g., visual words)

(it's a function that returns a single probability value)



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Naïve Bayes

Recall:

The posterior can be decomposed according to
Bayes' Rule

$$p(A|B) = \frac{\overset{\text{likelihood}}{p(B|A)} \overset{\text{prior}}{p(A)}}{\underset{\text{posterior}}{p(B)}}$$

In our context...

$$p(z|x_1, \dots, x_N) = \frac{p(x_1, \dots, x_N|z)p(z)}{p(x_1, \dots, x_N)}$$



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Naïve Bayes

The naive Bayes' classifier is solving this optimization

$$\hat{z} = \arg \max_{z \in \mathcal{Z}} p(z|\mathbf{X})$$

MAP (maximum a posteriori) estimate

$$\hat{z} = \arg \max_{z \in \mathcal{Z}} \frac{p(\mathbf{X}|z)p(z)}{p(\mathbf{X})}$$

Bayes' Rule

$$\hat{z} = \arg \max_{z \in \mathcal{Z}} p(\mathbf{X}|z)p(z)$$

Remove constants

To optimize this...we need to compute this



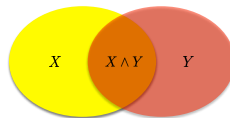
Compute the likelihood...

Naïve Bayes

A naive Bayes' classifier assumes all features are **conditionally independent**

$$\begin{aligned} p(\mathbf{x}_1, \dots, \mathbf{x}_N | \mathbf{z}) &= p(\mathbf{x}_1 | \mathbf{z}) p(\mathbf{x}_2, \dots, \mathbf{x}_N | \mathbf{z}) \\ &= p(\mathbf{x}_1 | \mathbf{z}) p(\mathbf{x}_2 | \mathbf{z}) p(\mathbf{x}_3, \dots, \mathbf{x}_N | \mathbf{z}) \\ &= p(\mathbf{x}_1 | \mathbf{z}) p(\mathbf{x}_2 | \mathbf{z}) \cdots p(\mathbf{x}_N | \mathbf{z}) \end{aligned}$$

Recall:



$$p(x, y) = p(x|y)p(y)$$



$$p(x, y) = p(x)p(y)$$

Naïve Bayes

To compute the MAP estimate

Given (1) a set of known parameters

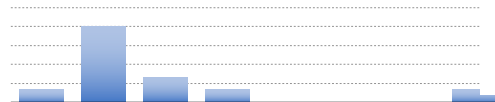
(2) observations

$$p(\mathbf{z}) \quad p(\mathbf{x}|\mathbf{z}) \quad \{x_1, x_2, \dots, x_N\}$$

Compute which $\mathbf{z}$ has the largest probability

$$\hat{\mathbf{z}} = \arg \max_{\mathbf{z} \in \mathcal{Z}} p(\mathbf{z}) \prod_n p(x_n|\mathbf{z})$$

Naïve Bayes



count	1	6	2	1	0	0	0	1
word	Tartan	robot	CHIMP	CMU	bio	soft	ankle	sensor
p(x z)	0.09	0.55	0.18	0.09	0.0	0.0	0.0	0.09

$$p(X|\mathbf{z}) = \prod_v p(x_v|\mathbf{z})^{c(w_v)}$$

$$= (0.09)^1 (0.55)^6 \dots (0.09)^1$$

Numbers get really small so use log probabilities

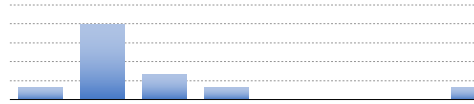
$$\log p(X|\mathbf{z} = \text{'grandchallenge'}) = -2.42 - 3.68 - 3.43 - 2.42 - 0.07 - 0.07 - 0.07 - 2.42 = -14.58$$

$$\log p(X|\mathbf{z} = \text{'softrobot'}) = -7.63 - 9.37 - 15.18 - 2.97 - 0.02 - 0.01 - 0.02 - 2.27 = -37.48$$

* typically add pseudo-counts (0.001)

** this is an example for computing the likelihood, need to multiply times **prior** to get posterior

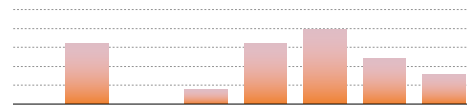
Naïve Bayes



count	1	6	2	1	0	0	0	1
word	Tartan	robot	CHIMP	CMU	bio	soft	ankle	sensor
p(x z)	0.09	0.55	0.18	0.09	0.0	0.0	0.0	0.09

$$\log p(X|z=\text{grand challenge}) = -14.58$$

$$\log p(X|z=\text{bio inspired}) = -37.48$$



count	0	4	0	1	4	5	3	2
word	Tartan	robot	CHIMP	CMU	bio	soft	ankle	sensor
p(x z)	0.0	0.21	0.0	0.05	0.21	0.26	0.16	0.11

$$\log p(X|z=\text{grand challenge}) = -94.06$$

$$\log p(X|z=\text{bio inspired}) = -32.41$$

* typically add pseudo-counts (0.001)

** this is an example for computing the likelihood, need to multiply times prior to get posterior



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Image Classification



(assume given set of discrete labels)
{dog, cat, truck, plane, ...}

→ cat



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Score function



→ class scores



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Linear Classifier

define a **score function**

data (histogram)

$$f(x_i, W, b) = Wx_i + b$$

class scores

“weights”

“bias vector”

“parameters”

Linear Classifier

Example with an image with 4 pixels, and 3 classes (cat/dog/ship)

Convert image to histogram representation



input image

0.2	-0.5	0.1	2.0
1.5	1.3	2.1	0.0
0	0.25	0.2	-0.3

W

56
231
24
2

x_i

+

1.1
3.2
-1.2

b

→

-96.8
437.9
61.95

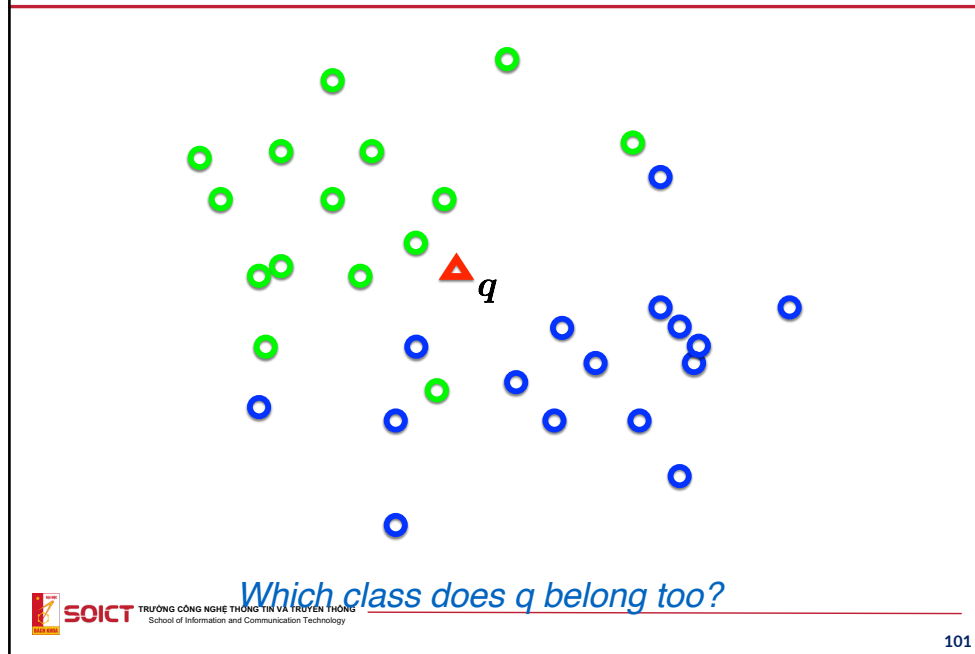
$f(x_i; W, b)$

cat score

dog score

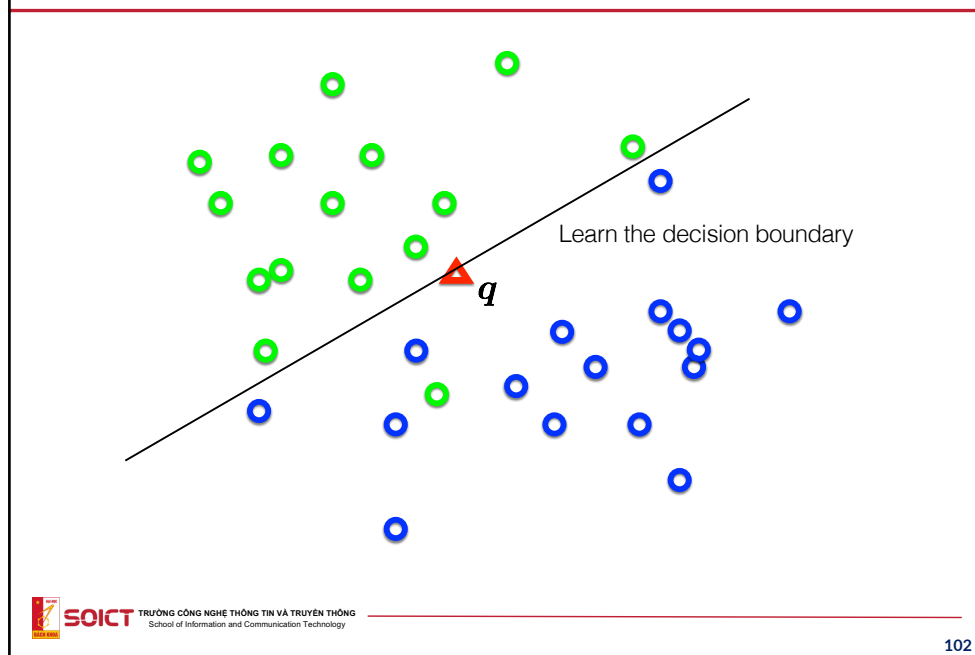
ship score

Distribution of data from two classes



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Distribution of data from two classes



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Remark

First we need to understand hyperplanes...



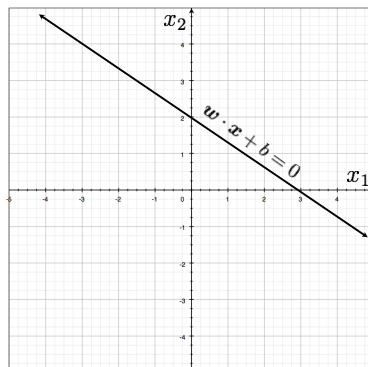
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Hyperplanes (lines) in 2D

$$w_1x_1 + w_2x_2 + b = 0$$



a line can be written as
dot product plus a bias

$$\mathbf{w} \cdot \mathbf{x} + b = 0$$

$$\mathbf{w} \in \mathcal{R}^2$$

another version, add a weight 1
and push the bias inside

$$\mathbf{w} \cdot \mathbf{x} = 0$$

$$\mathbf{w} \in \mathcal{R}^3$$



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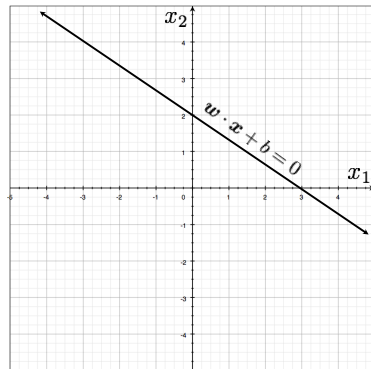
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Hyperplanes (lines) in 2D

$$\mathbf{w} \cdot \mathbf{x} + b = 0 \quad (\text{offset/bias outside}) \quad \mathbf{w} \cdot \mathbf{x} = 0 \quad (\text{offset/bias inside})$$

$$w_1x_1 + w_2x_2 + b = 0$$



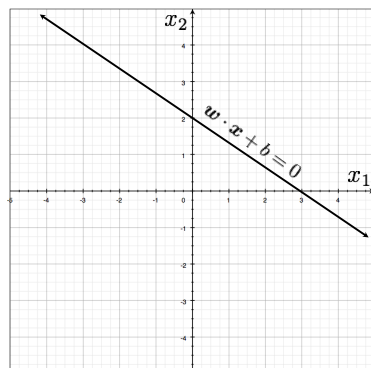
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Hyperplanes (lines) in 2D

$$\mathbf{w} \cdot \mathbf{x} + b = 0 \quad (\text{offset/bias outside}) \quad \mathbf{w} \cdot \mathbf{x} = 0 \quad (\text{offset/bias inside})$$

$$w_1x_1 + w_2x_2 + b = 0$$



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Important property:
Free to choose any normalization of w

The line

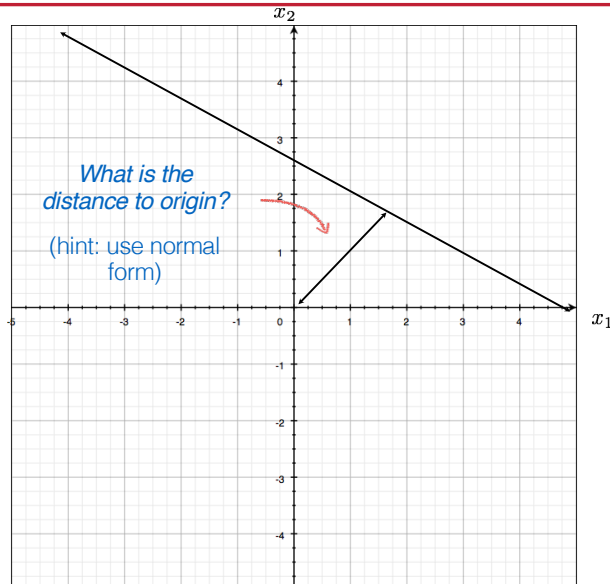
$$w_1x_1 + w_2x_2 + b = 0$$

and the line

$$\lambda(w_1x_1 + w_2x_2 + b) = 0$$

define the same line

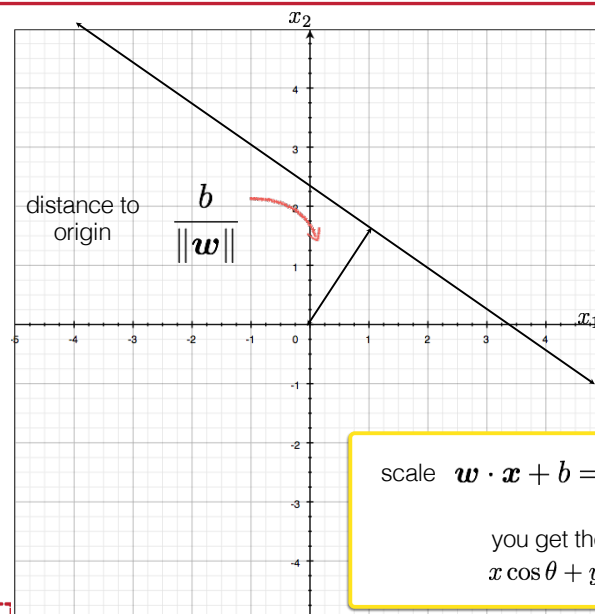
Hyperplanes (lines) in 2D



$$w \cdot x + b = 0$$

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Hyperplanes (lines) in 2D

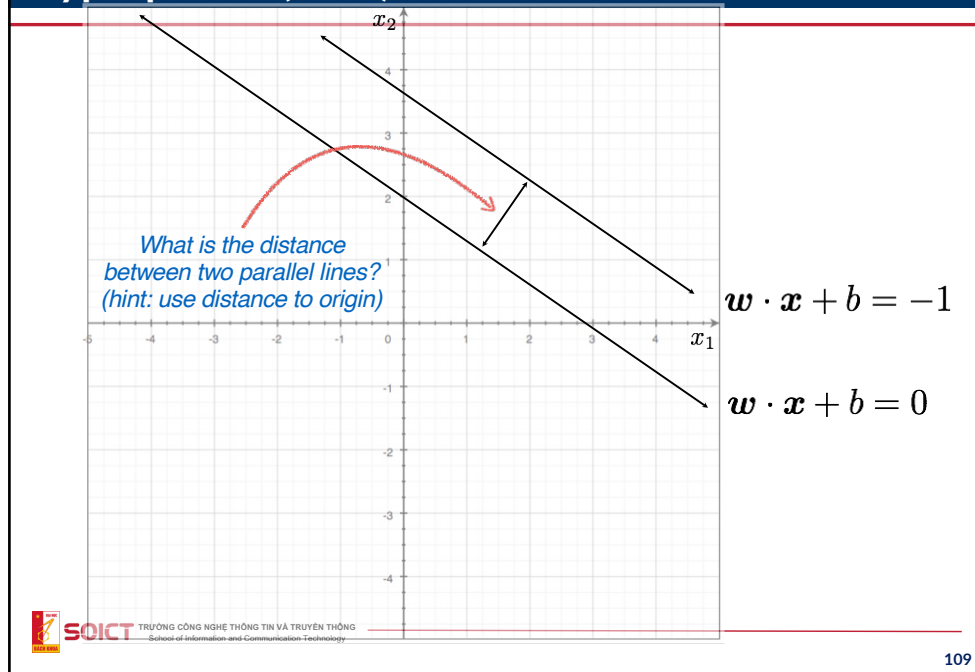


$$w \cdot x + b = 0$$

scale $w \cdot x + b = 0$ by $\frac{1}{\|w\|}$
you get the normal form
 $x \cos \theta + y \sin \theta = \rho$

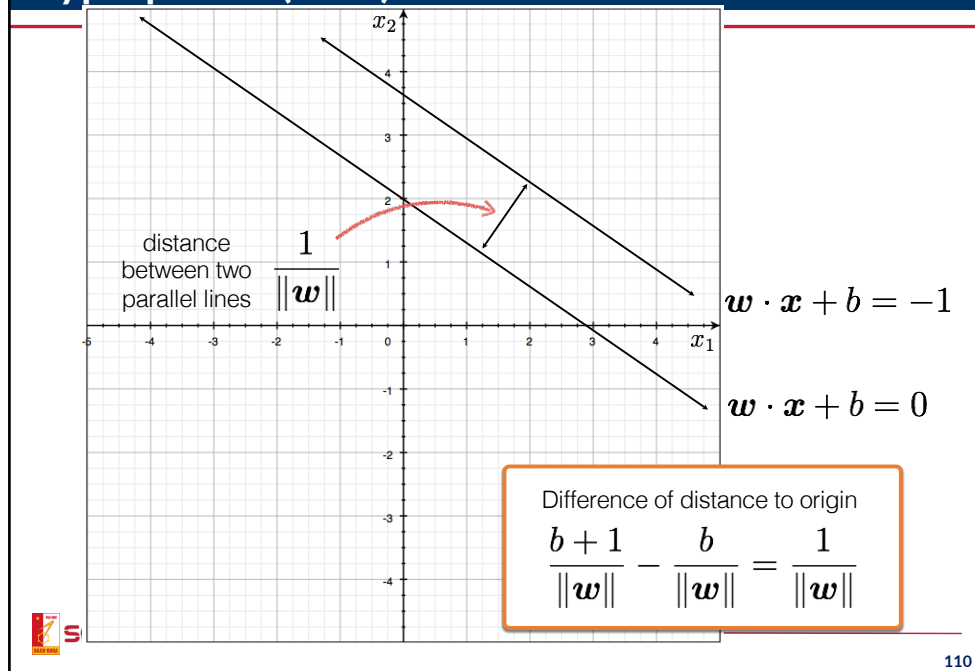
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Hyperplanes (lines) in 2D

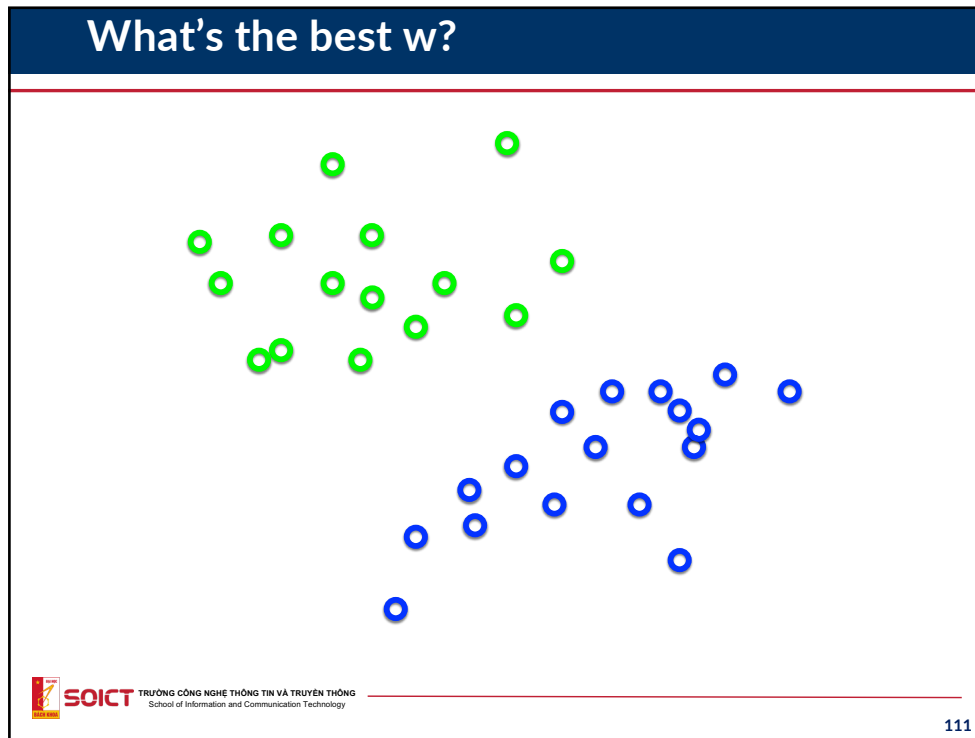


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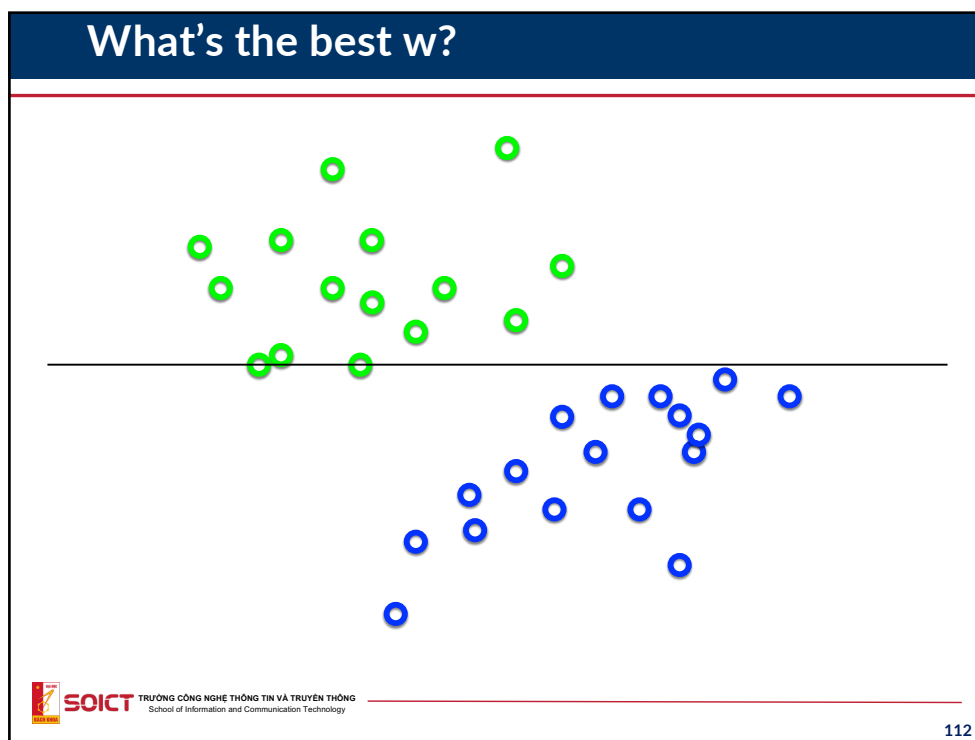
Hyperplanes (lines) in 2D



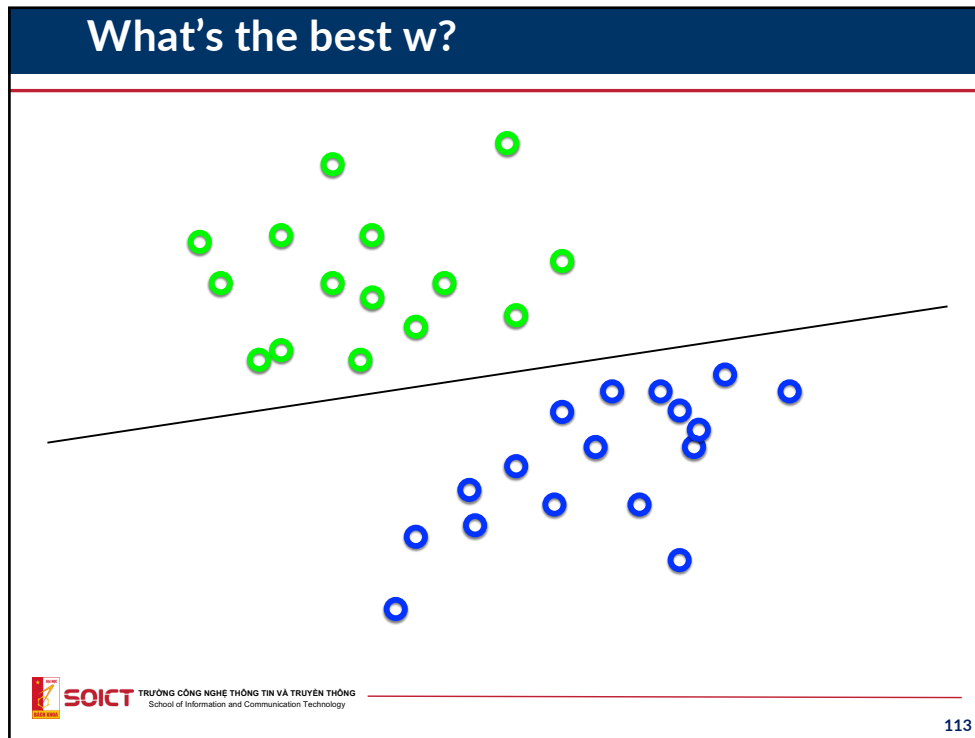
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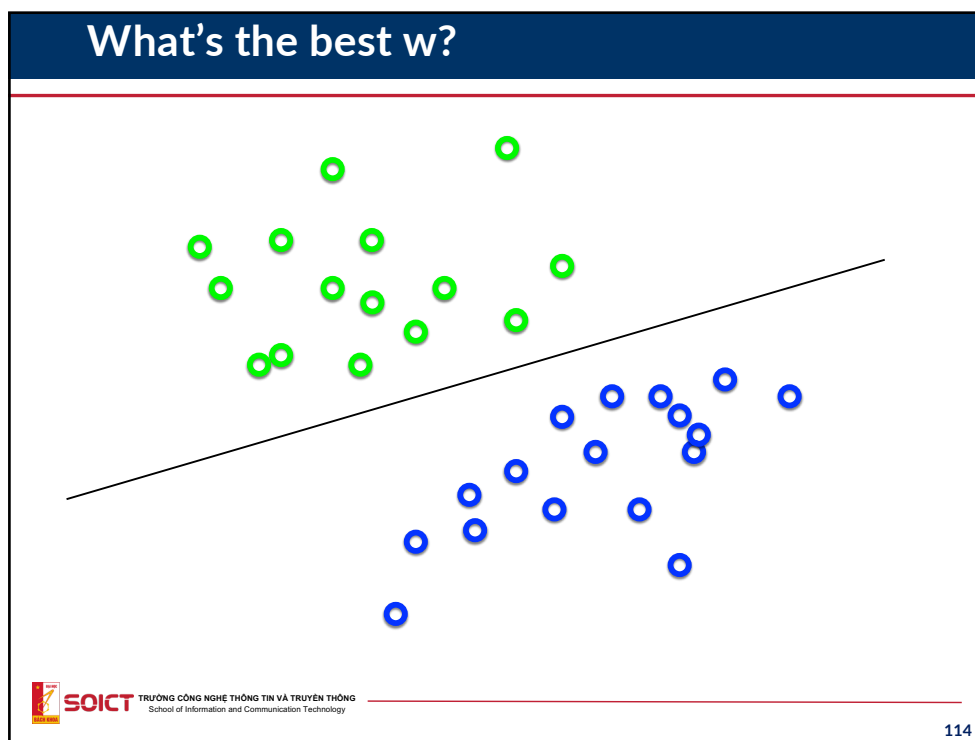
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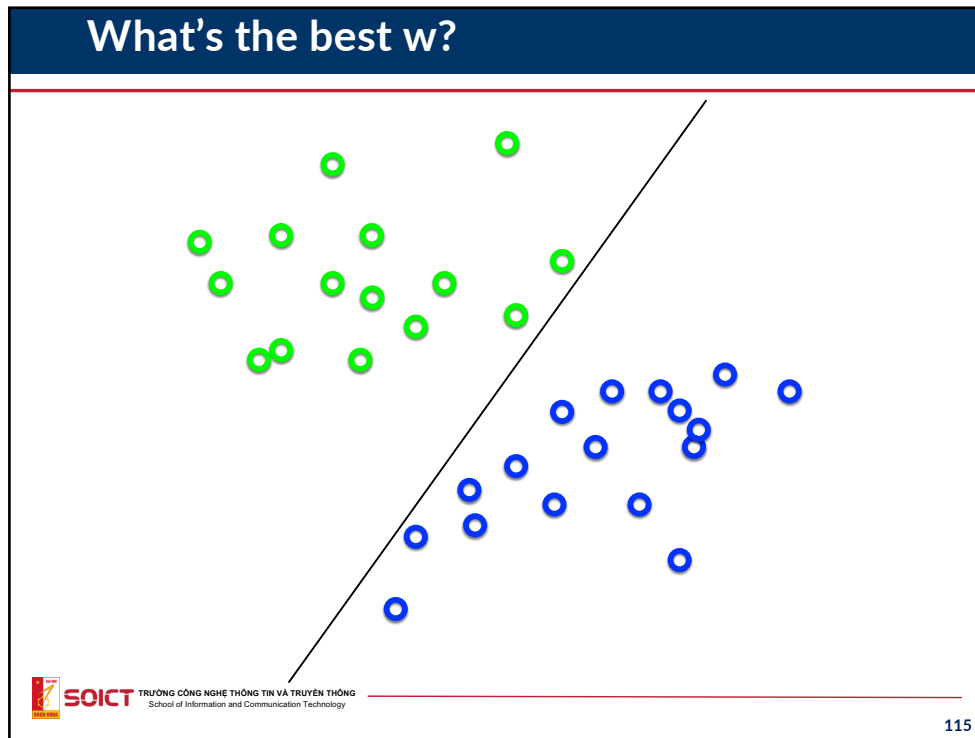
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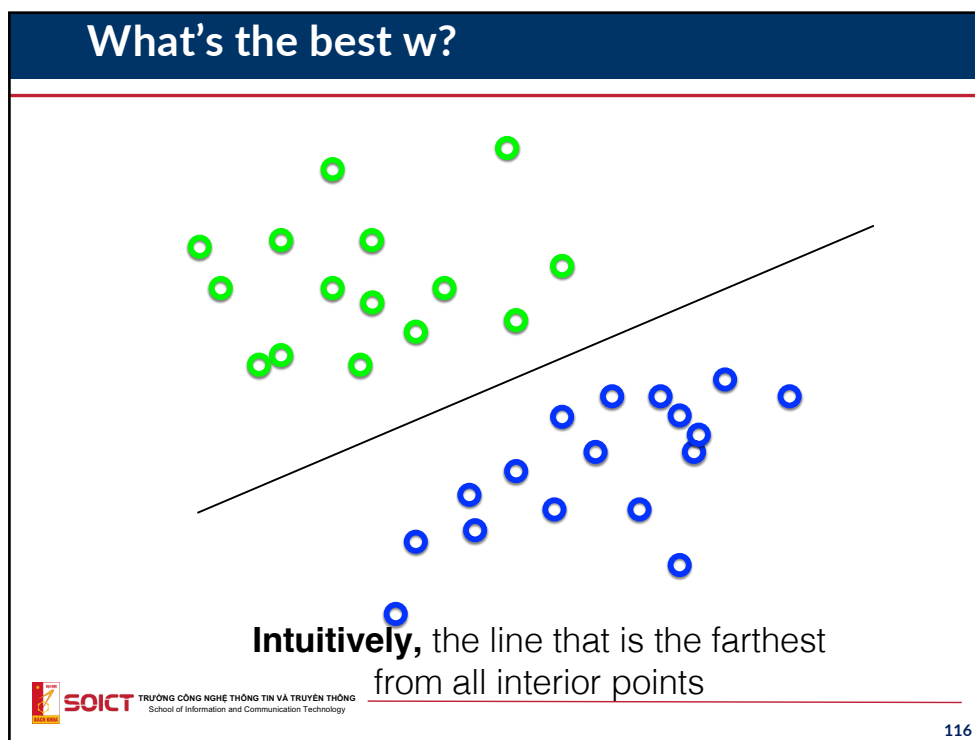
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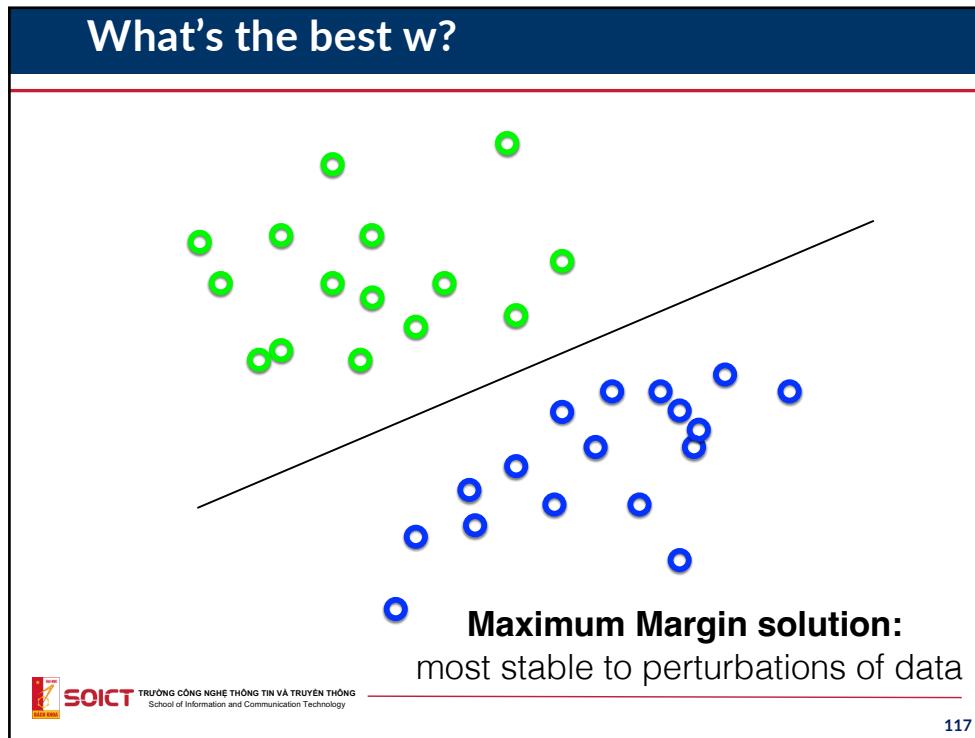


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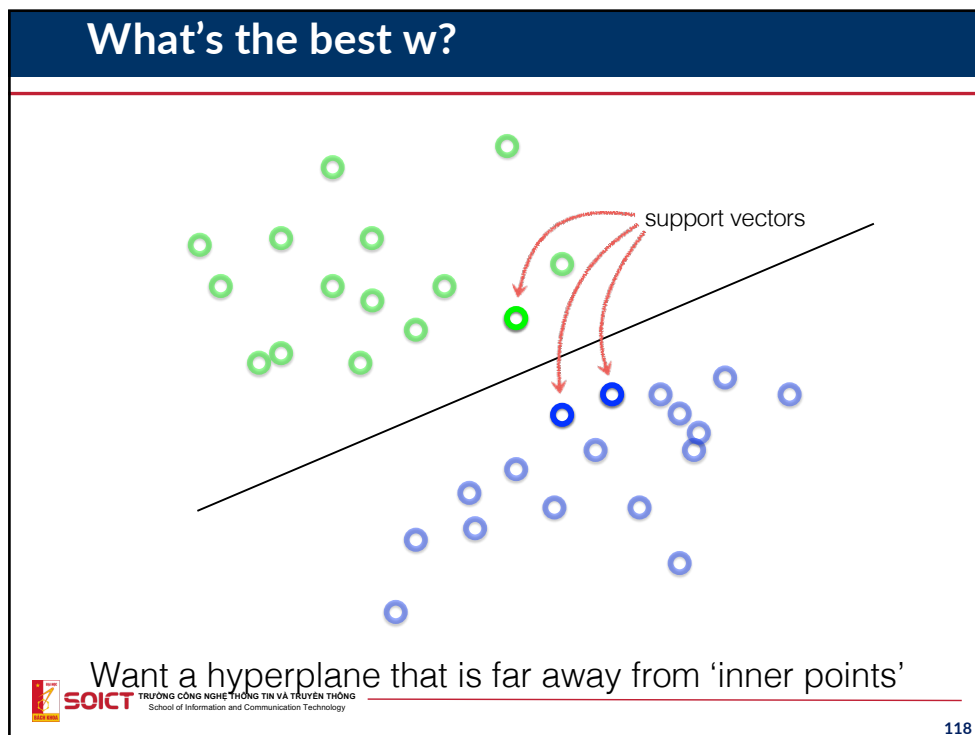
What's the best w ?



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What's the best w ?



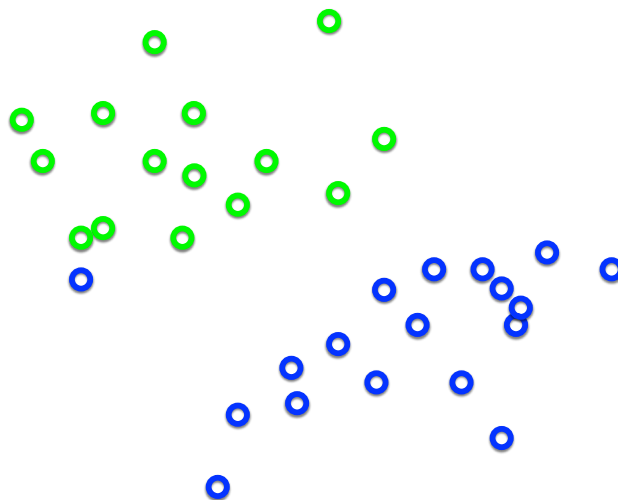
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Soft margin

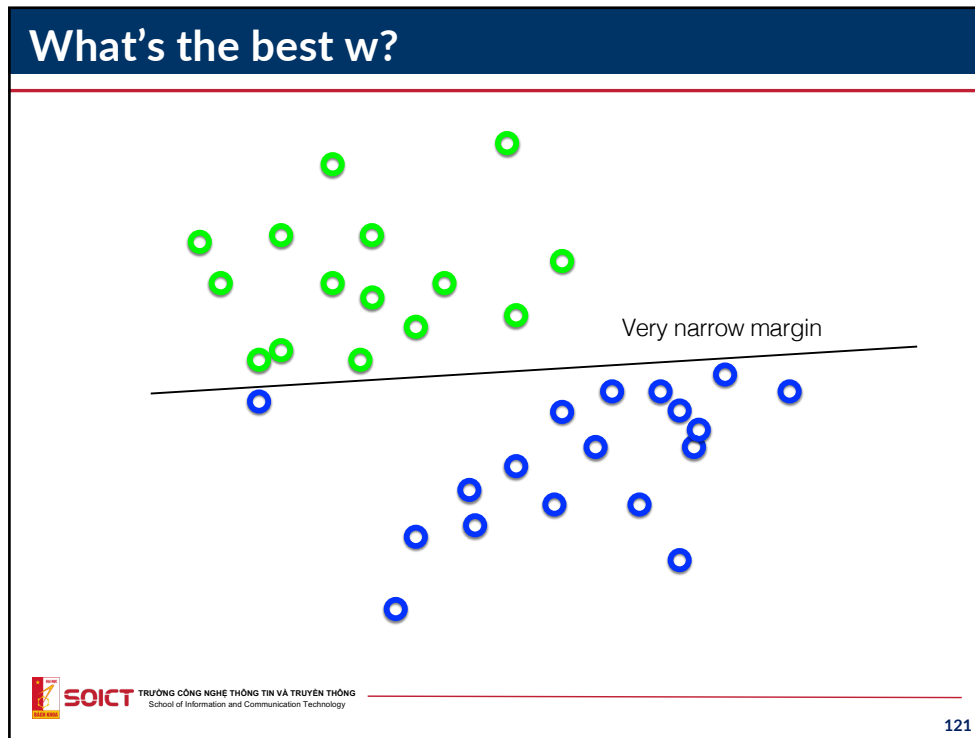
119

What's the best w ?



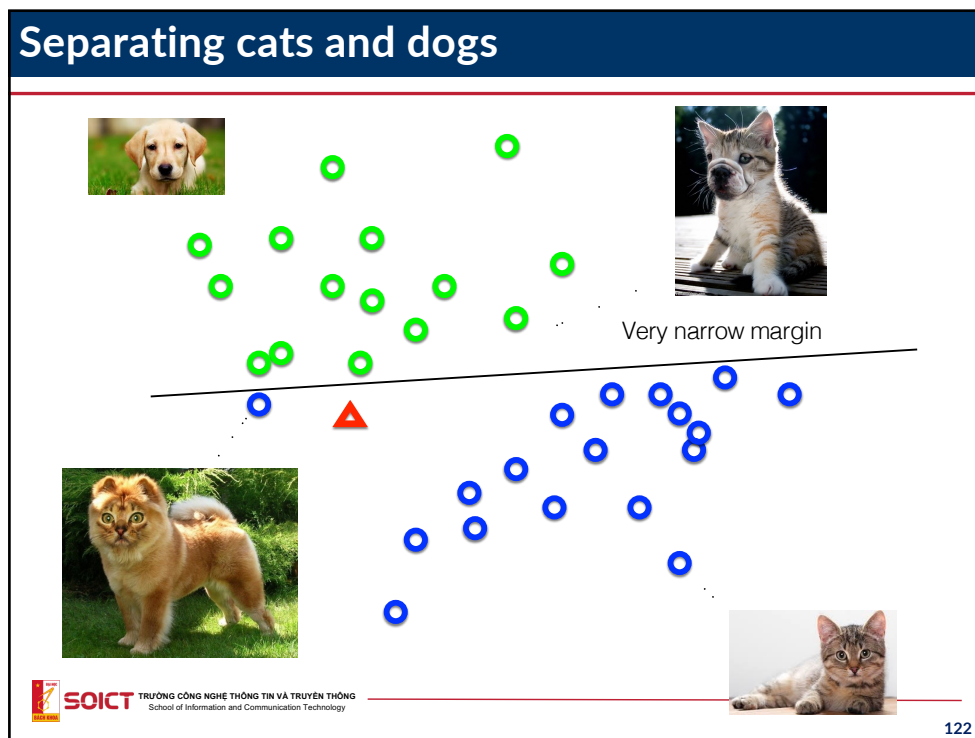
120

What's the best w ?



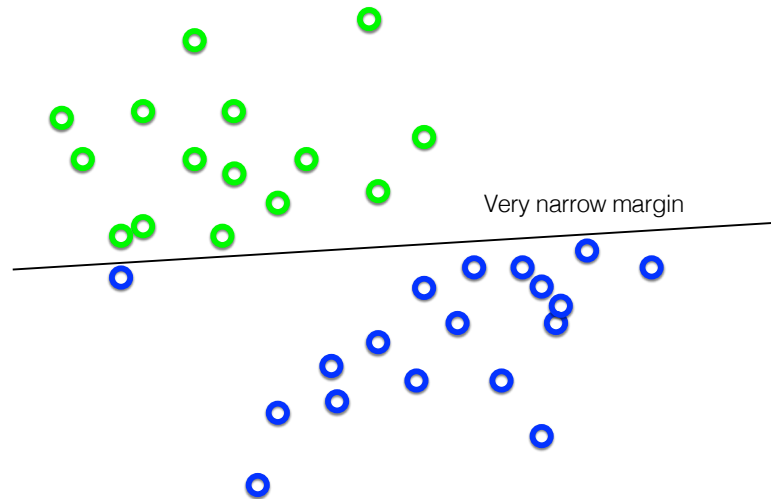
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Separating cats and dogs



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What's the best w ?



Intuitively, we should allow for some misclassification if we can get more robust classification

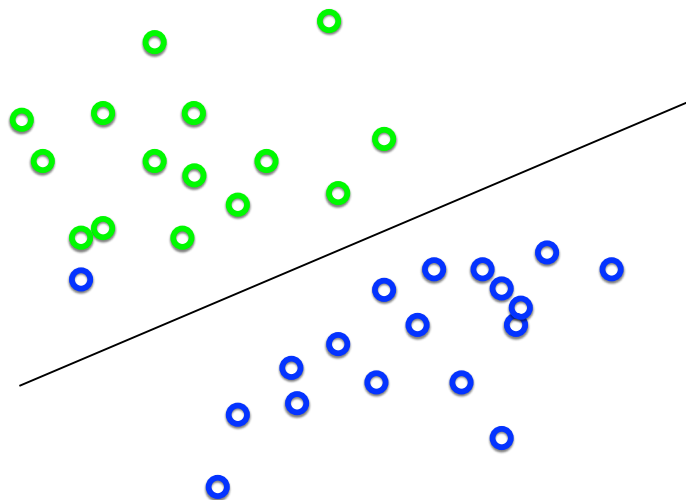


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What's the best w ?



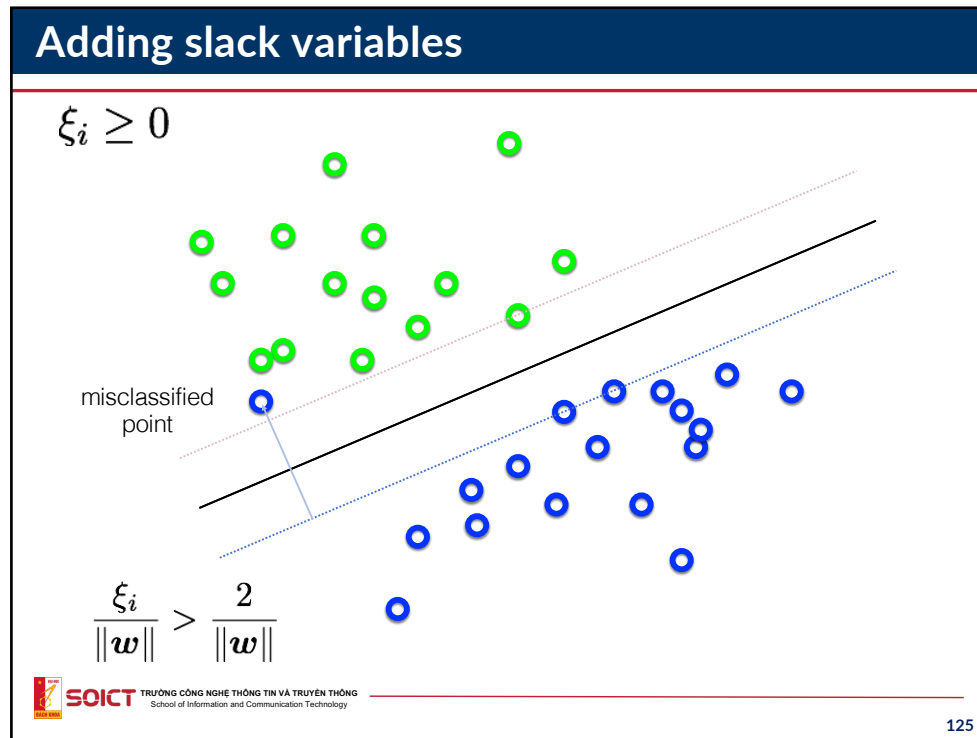
Trade-off between the MARGIN and the MISTAKES
(might be a better solution)



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Contents

- Overview of 'semantic vision'?
- Image classification
- Bag-of-words
 - Recall
 - Bag-of-feature
 - BOW for image classification
- Classification
 - K nearest neighbors
 - Naïve Bayes
 - Support vector machine
- Evaluation and hyperparameters selection

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Hyperparameters

- What is the best distance to use?
- What is the best value of k to use?
- i.e., how do we set the hyperparameters?
- Very problem-dependent
- Must try them all and see what works best



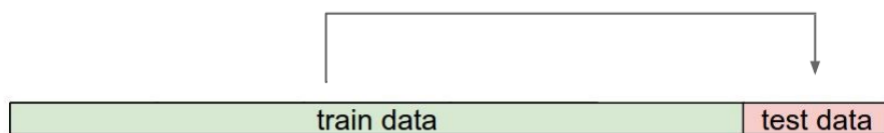
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Validation

Try out what hyperparameters work best on test set.



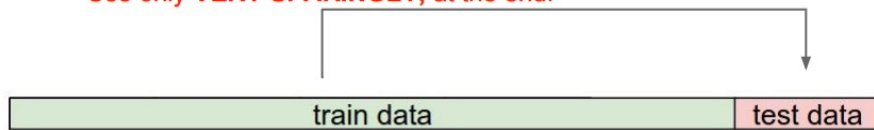
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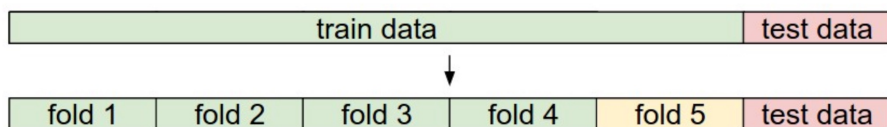
Validation

Trying out what hyperparameters work best on test set:
 Very bad idea. The test set is a proxy for the generalization performance!
 Use only **VERY SPARINGLY**, at the end.



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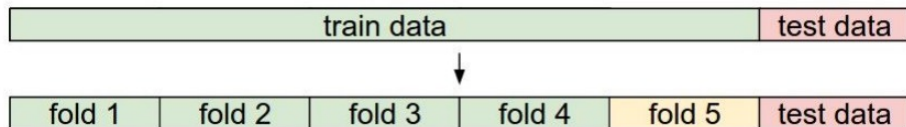
Validation



Validation data
 use to tune hyperparameters
 evaluate on test set ONCE at the end

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Cross-validation

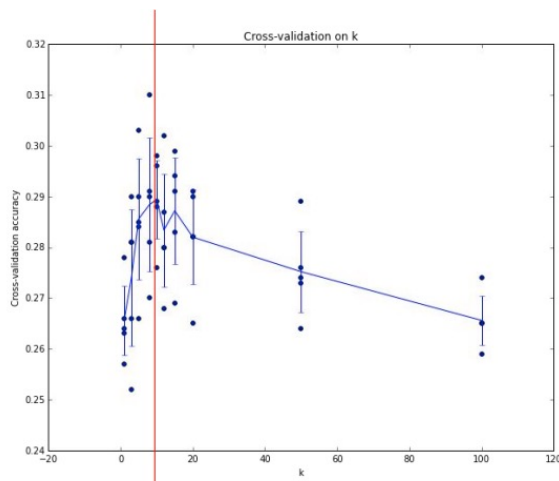


Cross-validation

cycle through the choice of which fold is the validation fold, average results.

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Cross-validation



Example of 5-fold cross-validation for the value of k .

Each point: single outcome.

The line goes through the mean, bars indicated standard deviation

(Seems that $k \approx 7$ works best for this data)

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How to pick hyperparameters?

- Methodology
 - Train and test
 - Train, validate, test
- Train for original model
- Validate to find hyperparameters
- Test to understand generalizability

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Confusion matrix

	Predicted 1	Predicted 0
True 1	true positive	false negative
True 0	false positive	true negative

	Predicted 1	Predicted 0
True 1	TP	FN
True 0	FP	TN

	Predicted 1	Predicted 0
True 1	hits	misses
True 0	false alarms	correct rejections

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

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Precision versus recall

- Accuracy
 - correct predictions / total predictions

- Precision:
 - how many of the object detections are correct?

$$precision = \frac{TP}{TP + FP}$$

- Recall:
 - how many of the ground truth objects can the model detect?
 - True Positive Rate (TPR)

$$recall = \frac{TP}{TP + FN}$$

- F1 score:
 - $2 * precision * recall / (precision + recall)$

References

Most of these slides were adapted from:

1. Ioannis Yannis, Gkioulekas (16-385 Computer Vision, Spring 2020, CMU)
2. Kristen Grauman (CS 376: Computer Vision, Spring 2018, The University of Texas at Austin)
3. Noah Snavely (Cornell University)
4. Fei-Fei Li (Stanford University)



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